06/09/2024

Math 4141

Coordinate Geometry
+
Differential calculus.

Differential Calculus
(Differentiation)

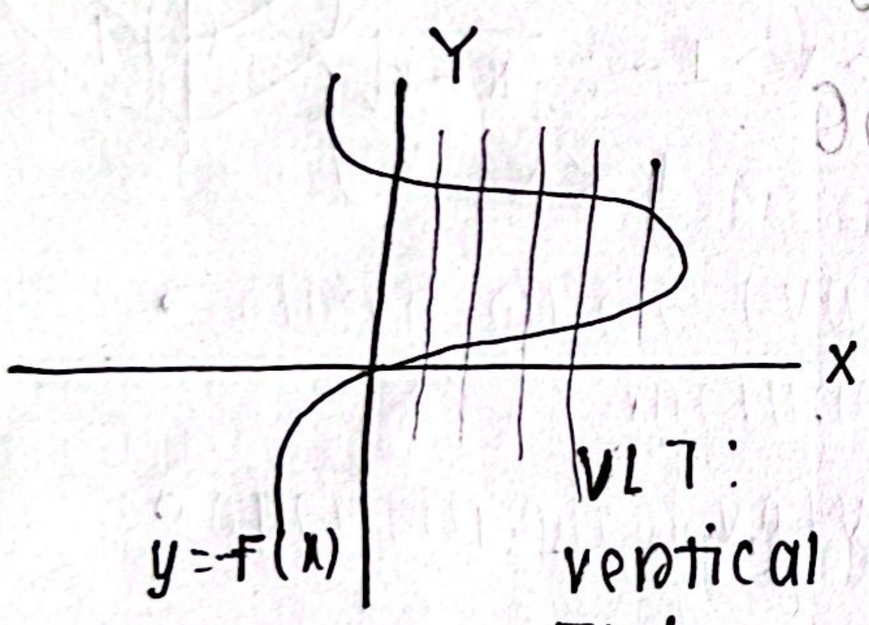
Google classroom :
Functions & Relation :

Function : For a distinct input it will have a distinct output.

$x \longrightarrow F(x)$

Input → Independent Variable

Output → Dependent variable → Function value.



VLT : Vertical Line Test

IF a single vertical lines intersects a function more than once than it is no longer be a function.

Function value can be determined by the height of the curve.

$x^2+y^2=16$ ⟶ Function containing more than one variable it is called implicit function.

$y=F(x)$ ⟶ the variable is solely described and function containing one variable is explicit.

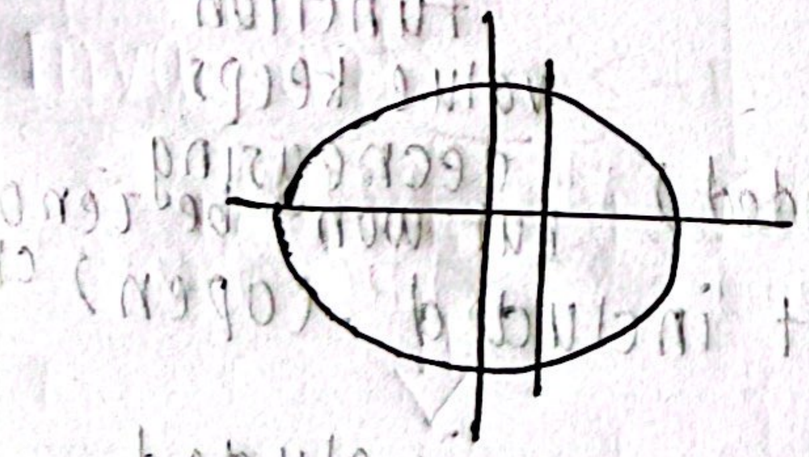
$x^2+y^2=16$ ⟶ Explicit function :

$\Rightarrow y^2=16-x^2$

$\Rightarrow y=\pm\sqrt{16-x^2}$

$\therefore y=F(x)=\pm\sqrt{16-x^2}$ → Implicit function

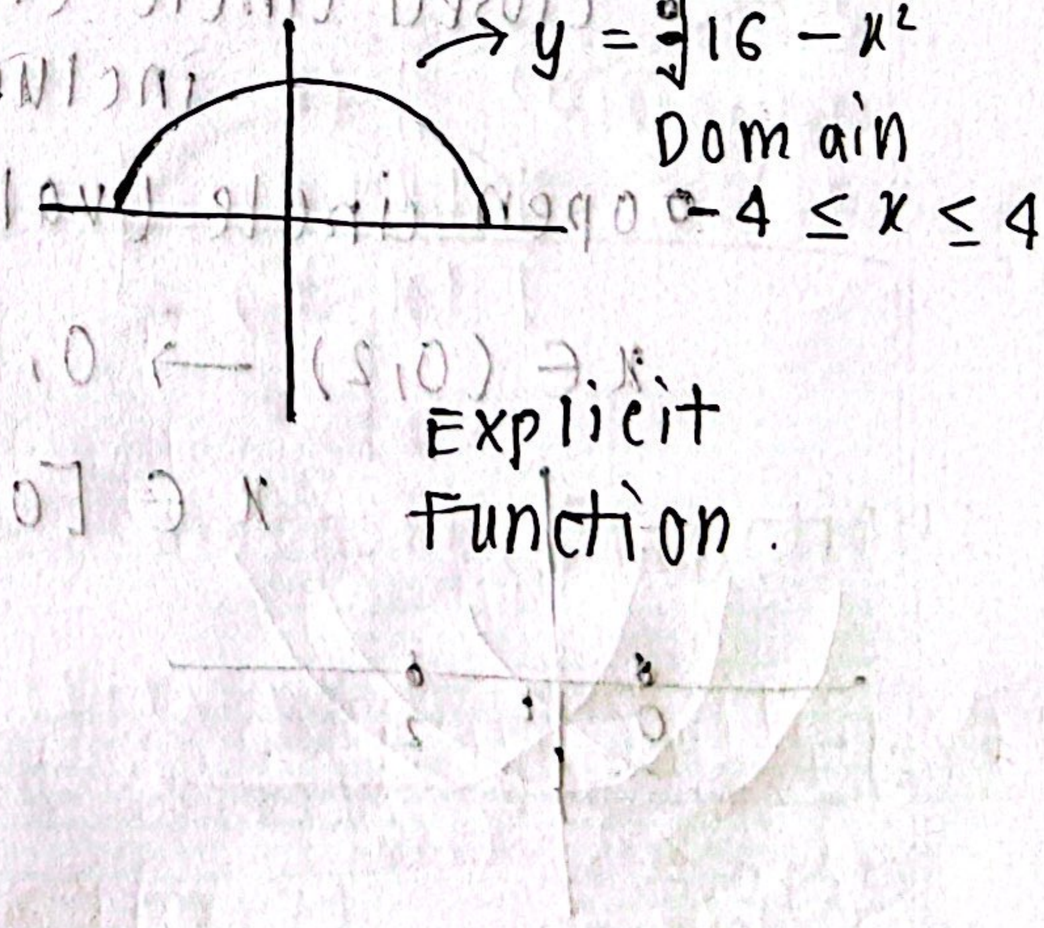
converting explicit function into implicit function (using either + or -)



Domain $-4\le x\le 4$

$x^2+y^2=16$

implicit function

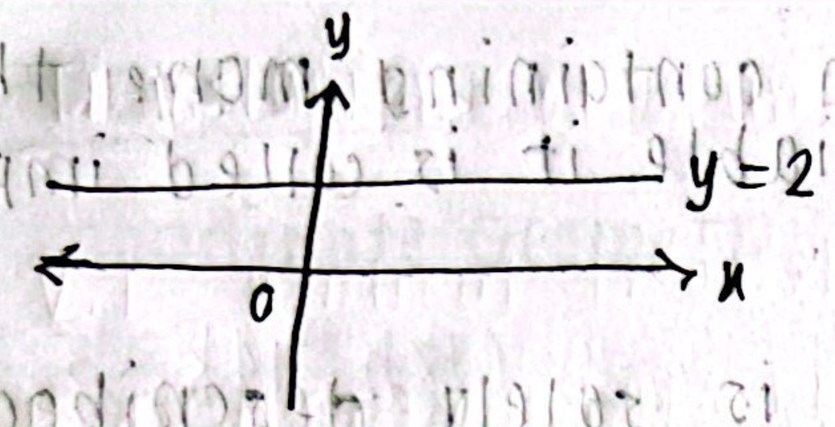


$y=\sqrt{16-x^2}$

Domain

$-4\le x\le 4$

Explicit function.



$x^3+3xy^2+y^3+2x^2y=17$

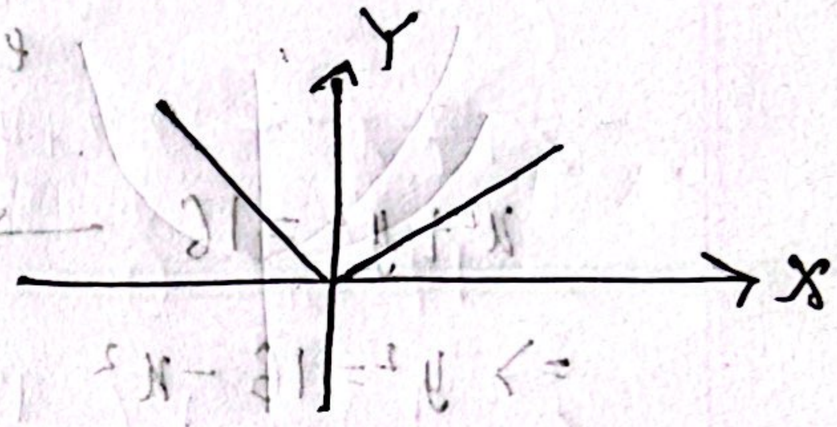
↳ can't be converted into implicit function.

↳ piece wise defined function.

value

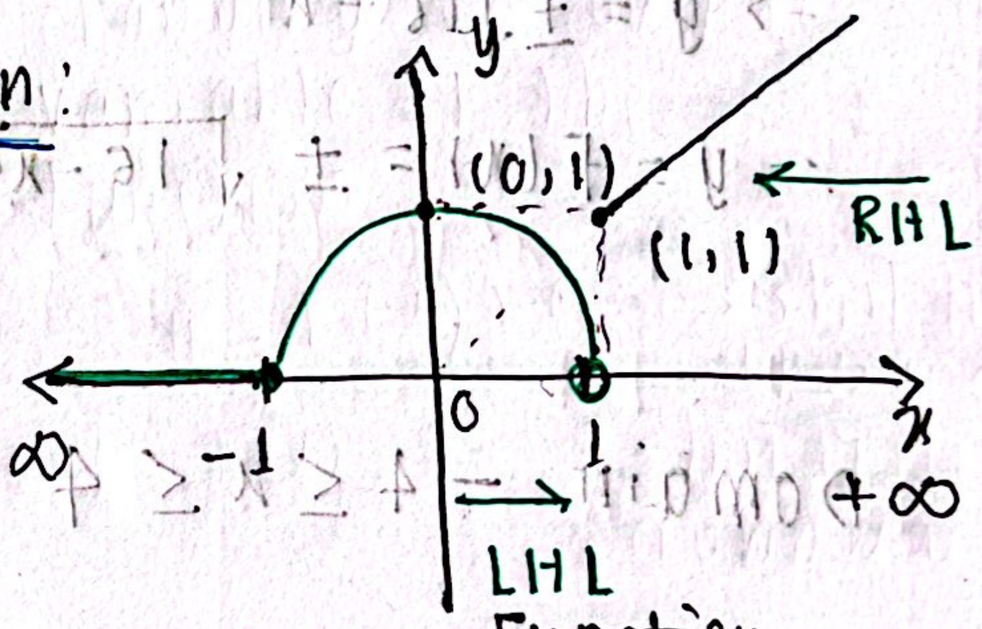
Absolute ~~Fu. zero~~ Function:

↳ $f(x) = |x| = \begin{cases} x\ ; x \geq 0 \\ -x\ ; x < 0 \end{cases}$ ←



Piecewise defined Function:

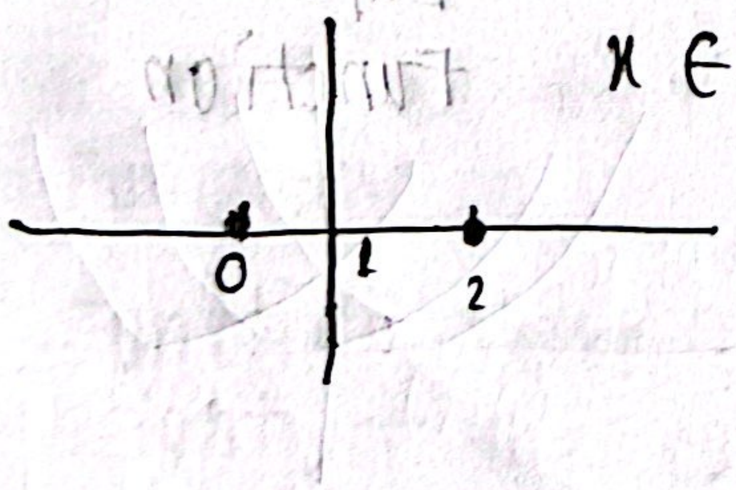
$$F(x) = \begin{cases} 0\ ; x \leq -1 \\ \sqrt{1-x^2}\ ; -1 < x < 1 \\ x\ ; x \geq 1 \end{cases}$$



- closed circle (value included)
- open circle (value excluded)

$x \in (0,2) \rightarrow$ 0, 2 are not included (open)

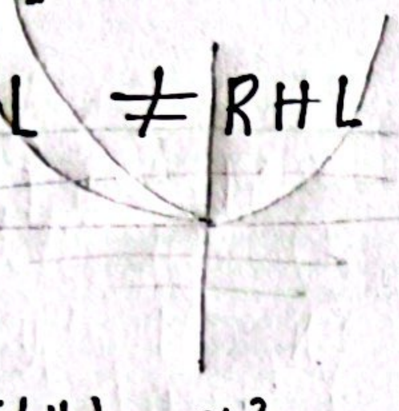
$x \in [0,2] \rightarrow$ 0, 2 are included (closed)



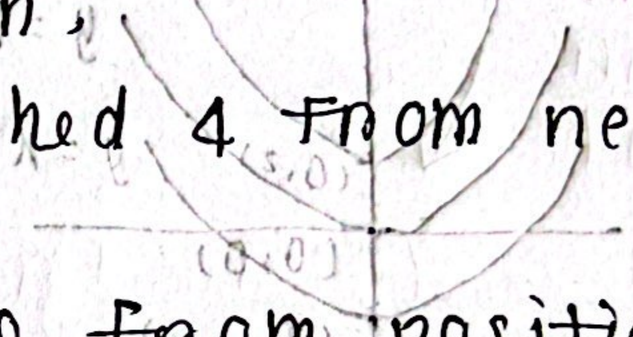
limit $f(x) = 0$ → limiting value ⟷ LHL
$x \to 1^-$
↳ x approaches to 1.

limit
$x \to 1^+$ $f(x) = 1$ → limiting value → RHL.

LHL ≠ RHL → If they do not match then the limit doesn't exist at that given point.

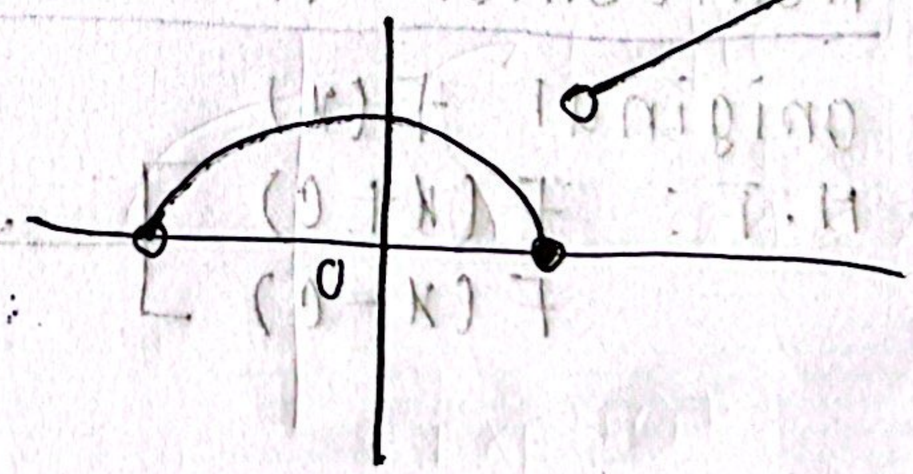


$F(x) = x^2$ when 1.9999 approaching towards two from negative direction, the function approached 4 from negative direction.



When x approaches towards two from positive direction, the function approaches 4 from positive direction.

$$F(x) = \begin{cases} 0, & x \le -1 \\ \sqrt{1-x^2}, & -1 < x \le 1 \\ x, & x > 1 \end{cases}$$



* composition of Function / composite Function:

composite Function:

$F \circ g$, $g \circ F$

$f(x) = x^2$

$g(x) = 3x + 7$

$F \circ g = F(g(x))$
$= F(3x+7)$
$= (3x+7)^2$

$g \circ F = g(F(x))$
$= g(x^2)$
$= 3x^2 + 7$.

Translation of function

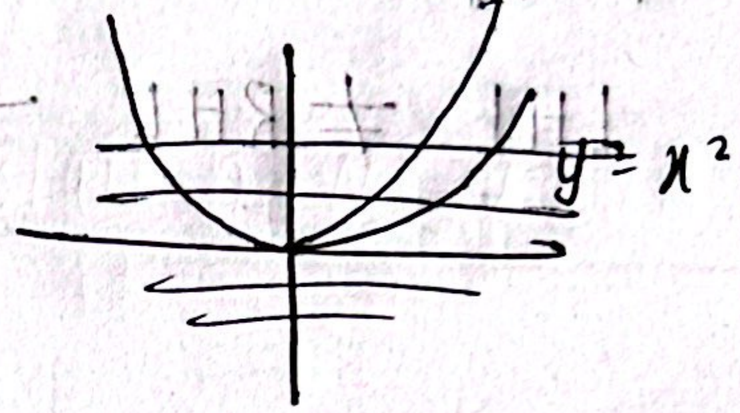
V.T : Vertical Translation / shifting

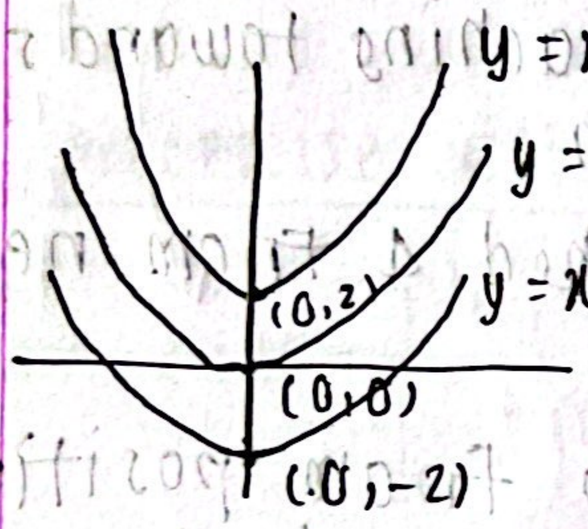
H.T : Horizontal Translation / shifting.

~~F(x)~~ Vertical Translation:

original position f(x)

V.T : $F(x)+C$, $F(x)-C$ → changing the output variable of the function





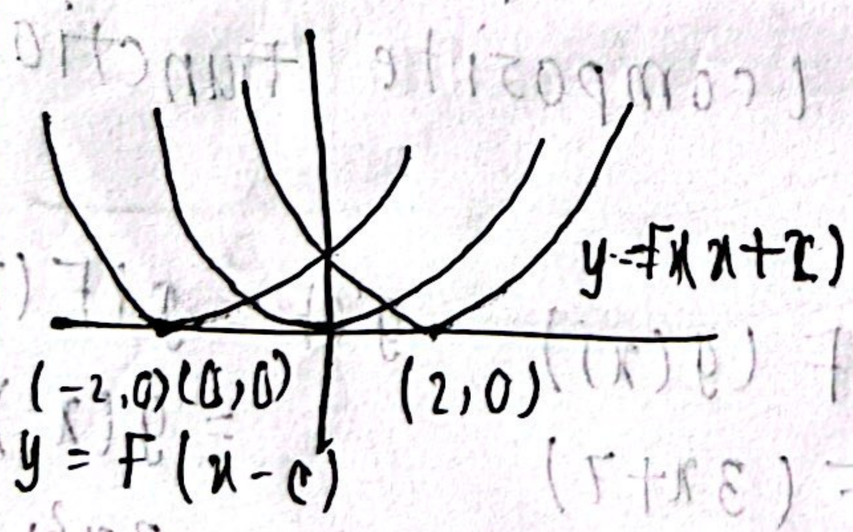
$y=F(x)+C$ → shifting upward

$y=F(x)-C$ → shifting downward.

Horizontal Translation:

original $F(x)$

H.T : $F(x+C)$, $F(x-C)$ → changing the input variable of the Function.



$y=F(x+C)$ → shifting ~~n i g~~ forward

$y=F(x-C)$ → shifting backward.

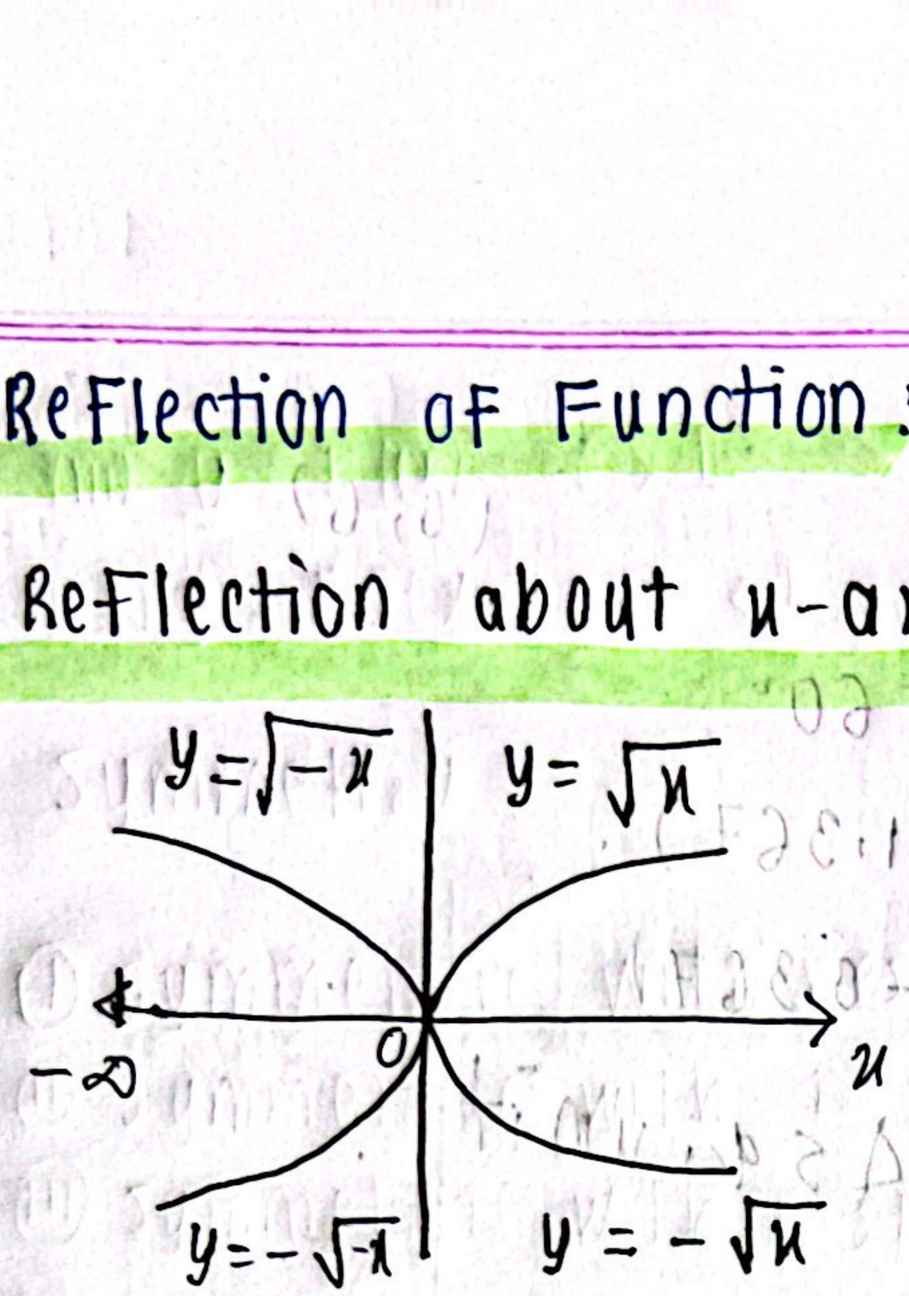
Next class: Reflection of function.

Refenence book :

13/09/2024

Reflection of Function:

Reflection about x-axis:



$y = \sqrt{-x}$ is possible if the domain remains within 0 to $-\infty$.

$x \longrightarrow (-x)$ (Reflection About y-axis)

$F(x) \longrightarrow -F(x)$ (Reflection About x-axis)

Stretching & compression of Function:

- compresses & stretches along vertical y-axis.
- compresses & stretches along horizontal x-axis.

* $y = F(x)$

(i) $y = cF(x)$; $c > 1$

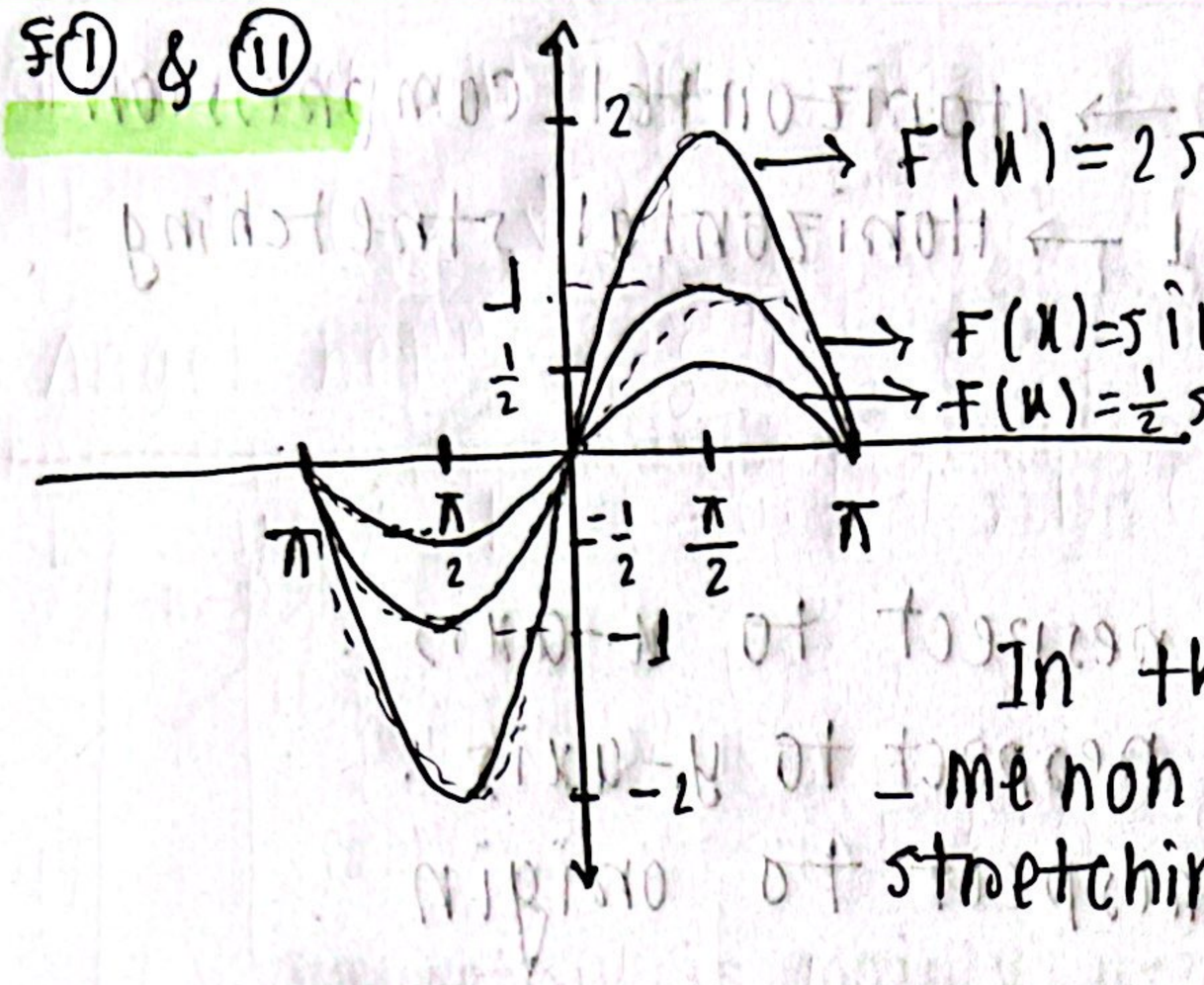
(ii) $y = cF(x)$; $0 < c < 1$ - Fraction.

(iii) $y = F(cx)$; $c > 1$

(iv) $y = F(cx)$; $0 < c < 1$.

In the first two the input is multiplied by a constant whereas the last two ~~input is~~ the function is multiplied by the constant.

① & ②



→ when we are multiplying the function with a constant it will stretch the graph.

In this graph this phenomenon describes vertical stretching.

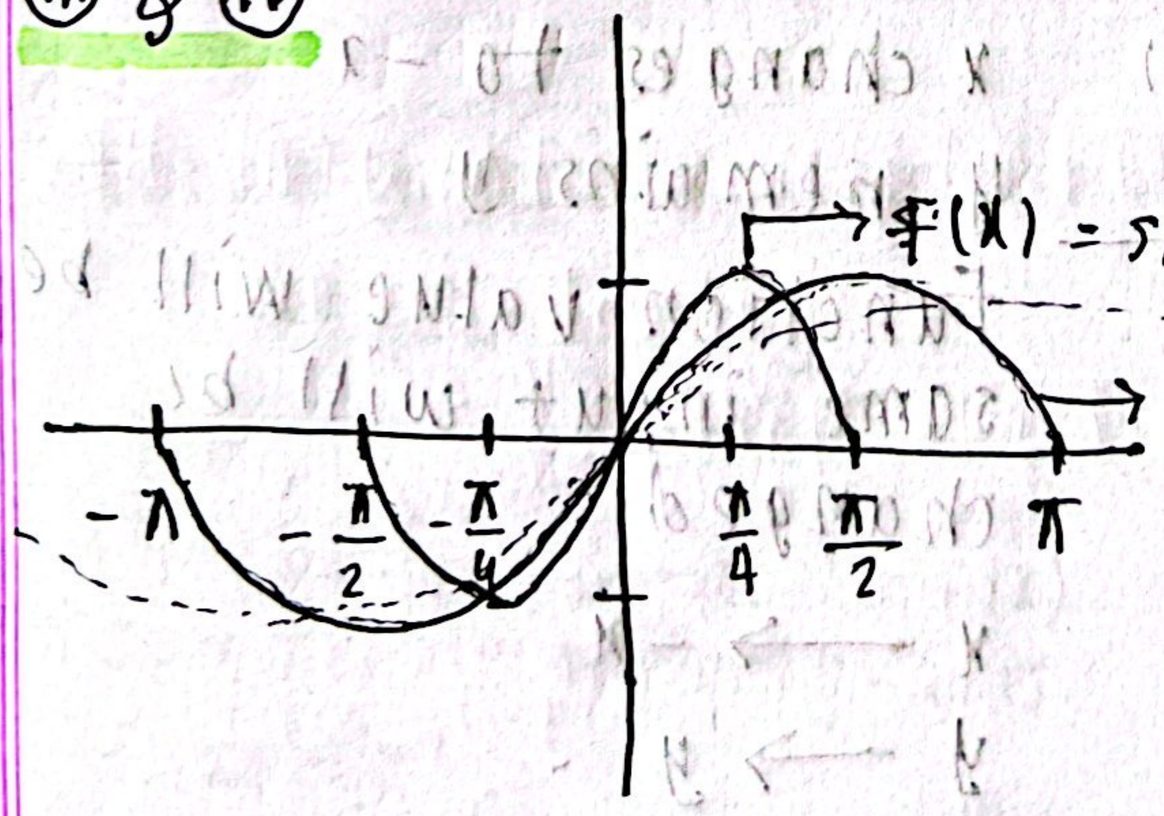
- When we multiplied $\frac{1}{2}$ with the function it described vertical compression.

(I) $y = cF(x)$; $c > 1$ ⟶ Vertical stretching.

(II) $y = cF(x)$; $0 < c < 1$ ⟶ Vertical compression by a factor $1/c$.

$\frac{1}{c}$ = how much time compression will take place.

③ & ④



④

$F(x) = \sin\frac{1}{2}x$

↳ Horizontal stretching

Horizontal compression 'c' times

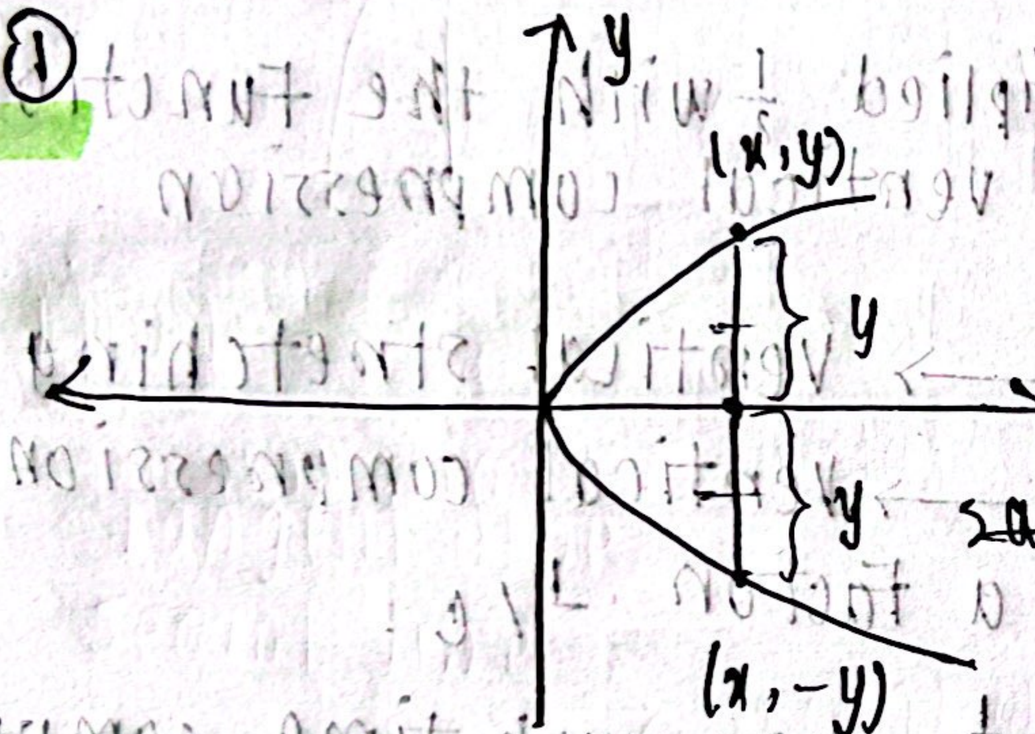
Horizontal stretching '$\frac{1}{c}$' times.

(iii) $y = f(cx)$; $c > 1$ → Horizontal compression

(iv) $y = f(cx)$; $0 < c < 1$ → Horizontal stretching.

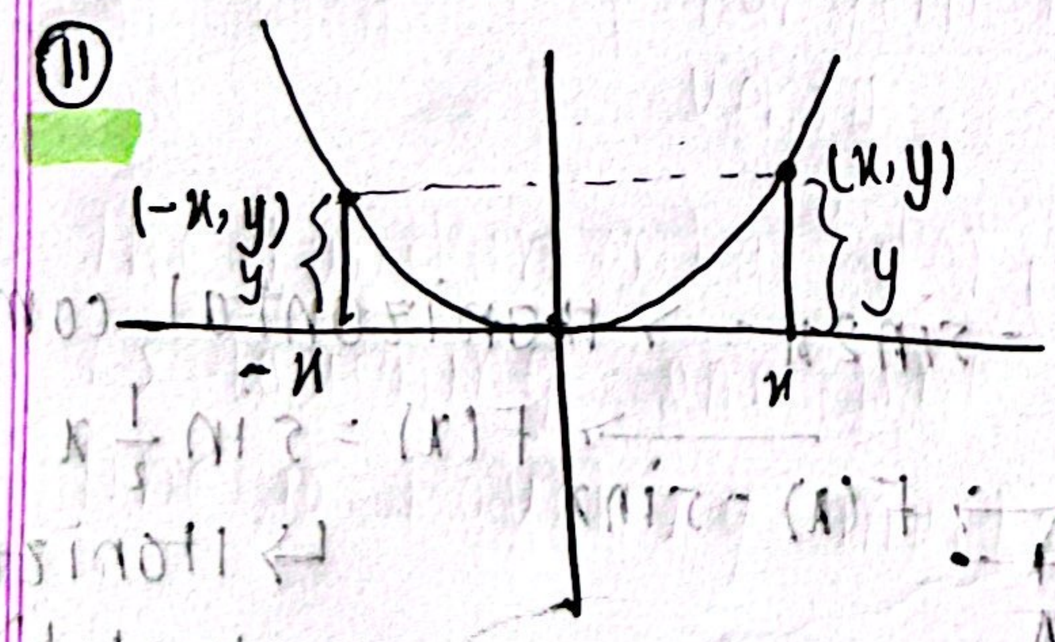
Symmetry

① symmetric with respect to x-axis.
② symmetric with respect to y-axis.
③ symmetric with respect to origin.

① 

x remains x
y changes to -y. change
function will ~~change~~ ~~same~~ & input value will change. remain same

$x \longrightarrow x$
$y \longrightarrow -y$.

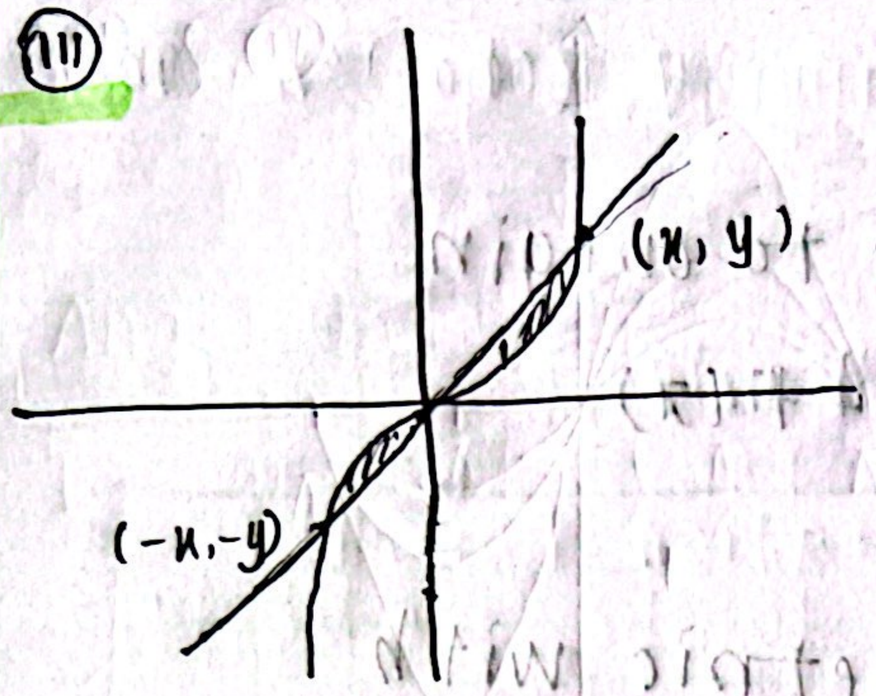
② 

x changes to -x
y remains y
function value will be same, input will be changed

$x \longrightarrow -x$
$y \longrightarrow y$.

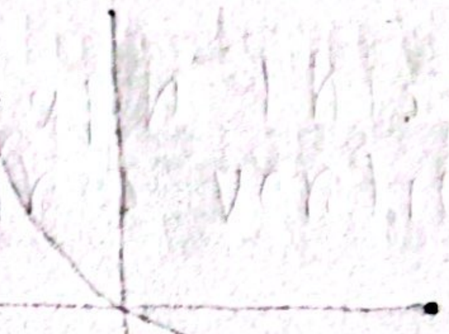
(ii) & (iii)

(iii)



x becomes $-x$
y becomes $+y$
both inputs & function values will change its sign.

$x \longrightarrow -x$
$y \longrightarrow -y$



Symmetry is mainly used in integral function.

Even & Odd function:

Two methods can be used to determine even & odd functions:

① Algebraic
② Graphical

Algebraic: $F(-x) = F(x) \longrightarrow$ even function.
$F(-x) = -F(x) \longrightarrow$ odd function.

Even Function:

$F(-x) = F(x)$

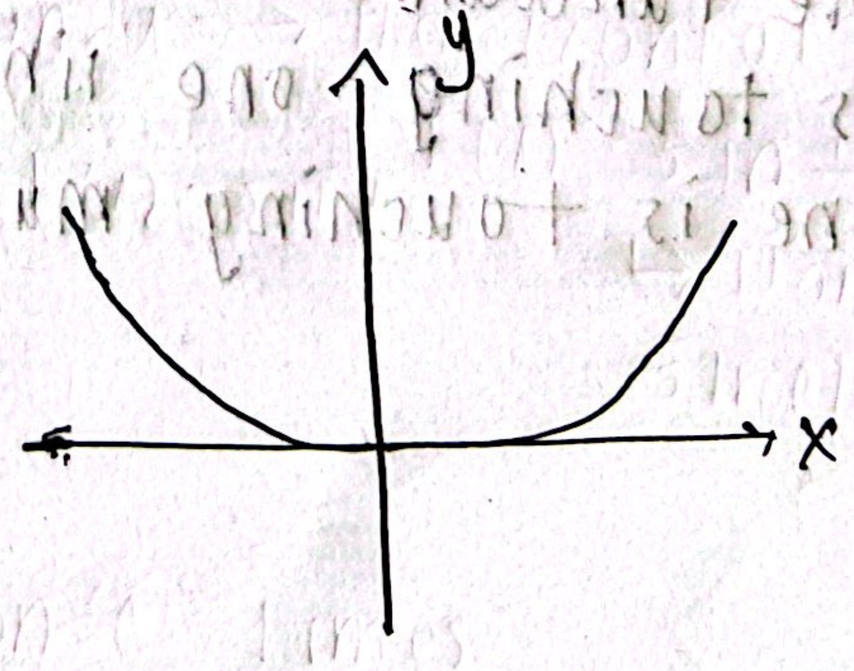
$F(x) = \cos x$

$F(-x) = \cos(-x) = \cos x = F(x)$

*symmetric w.r.t y-axis.

$F(x) = x^2$

$F(-x) = (-x)^2 = x^2 = F(x)$

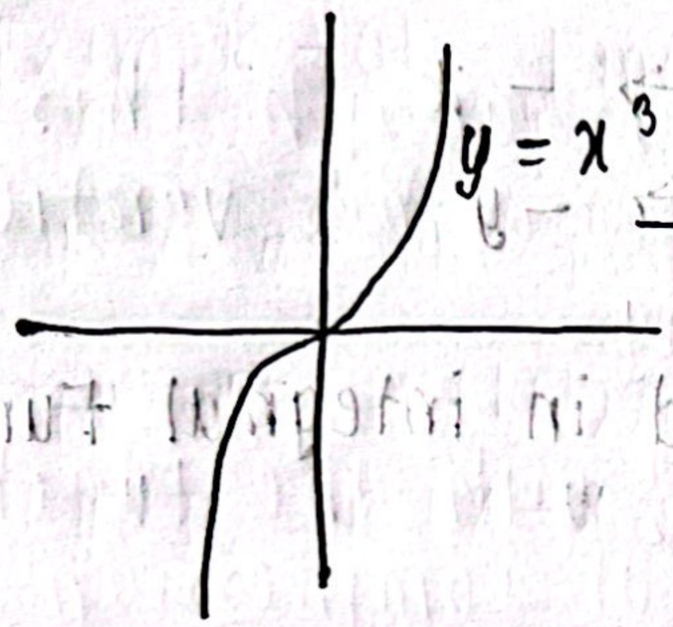


Odd function:

$f(-x) = -f(x)$

symmetric with respect to origin.

$f(-x) = \sin(-x) = -\sin x = -f(x)$



$\Longrightarrow$ symmetric with respect to origin.

* Differentiation → tangent line problems.

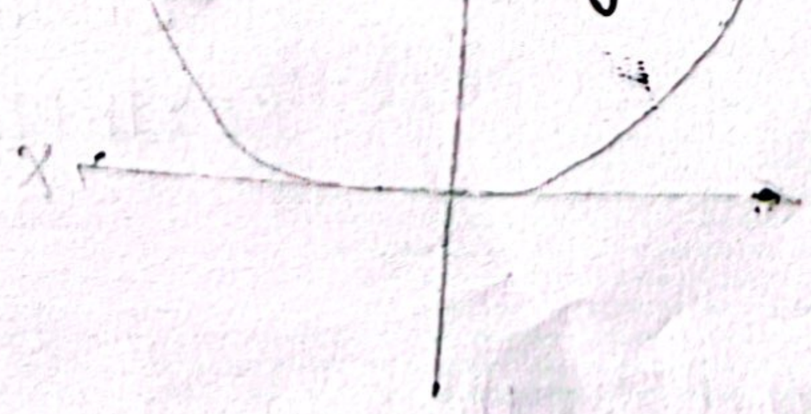
Limit

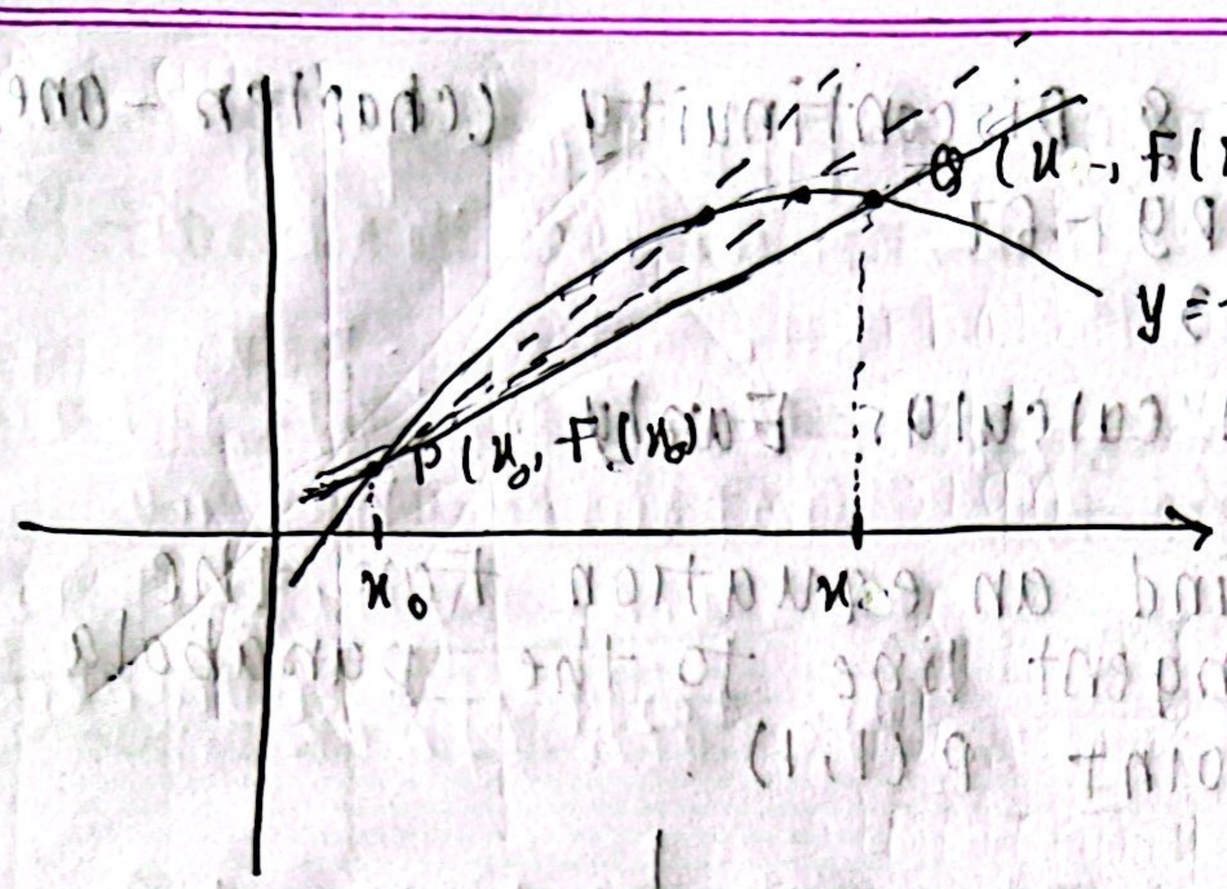
How limit is related to ~~teh~~ tangent line.

Tangent line

Secant line → secant line can approach to be a tangent line but tangent line cannot do the vice versa.

The fundamental definition of tangent line is touching one line & that of secant line is touching multiple lines.





For this purpose we need to apply the point slope form.

Straight line :

Slope form : ① Point slope form : $y - y_1 = m(x - x_0)$

② Slope intercept form : $y = mx + c$

③ Intercept form : $\frac{x}{a} + \frac{y}{b} = 1$.

Slope of PQ $\equiv m_{sec} = \frac{F(x) - F(x_0)}{x - x_0}$

* When x approaches the point x_0, Q approaches the point P. And gradually secand line approaches the tangent line.

slope of the tangent line :

slope of the tangent line at P :

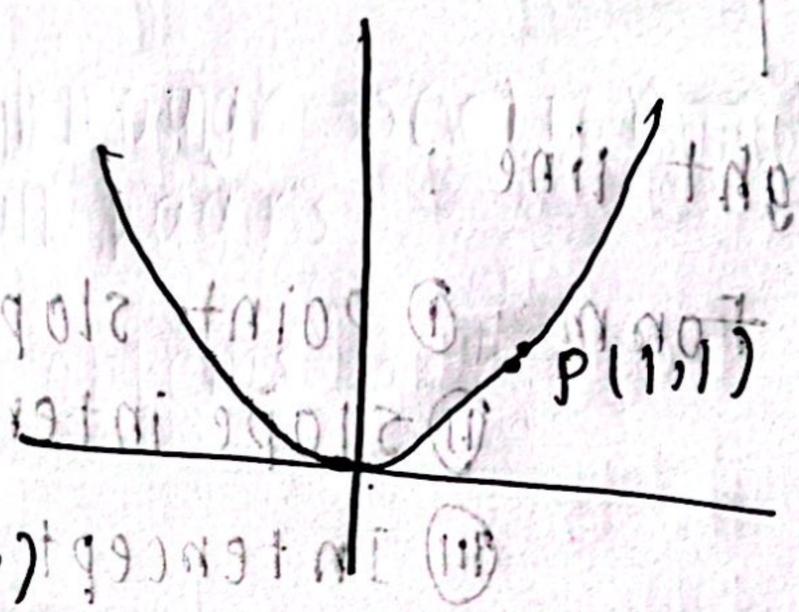
$$m_{tan} = \lim_{x \to x_0} (m_{sec})$$

$$= \lim_{x \to x_0} \frac{F(x) - F(x_0)}{x - x_0}$$

$$\therefore m_{tan} = \lim_{x \to x_0} \frac{F(x) - F(x_0)}{x - x_0}$$

Refenance: Limit & Discontinuity (chapten - one)
PdF pg - 67
pg - 43
Anton calculus Eanly

Example - 01: Find an equation for the tangent line to the parabola $y = x^2$ at the point $P(1,1)$.



$\Rightarrow$ $y = x^2$ $P(1,1)$

$y - y_0 = m(x - x_0)$

$m_{tan x} = \lim_{x \to 1} m_{sec}$

$= \lim_{x \to 1} \frac{f(x) - f(x_0)}{x - x_0}$

$= \lim_{x \to 1} \frac{x^2 - 1^2}{x - 1} = \lim_{x \to 1} (x+1) = 2$

problem: $f(x) = \frac{x^2 - 4}{x - 2} = \frac{p(x)}{q(x)}$

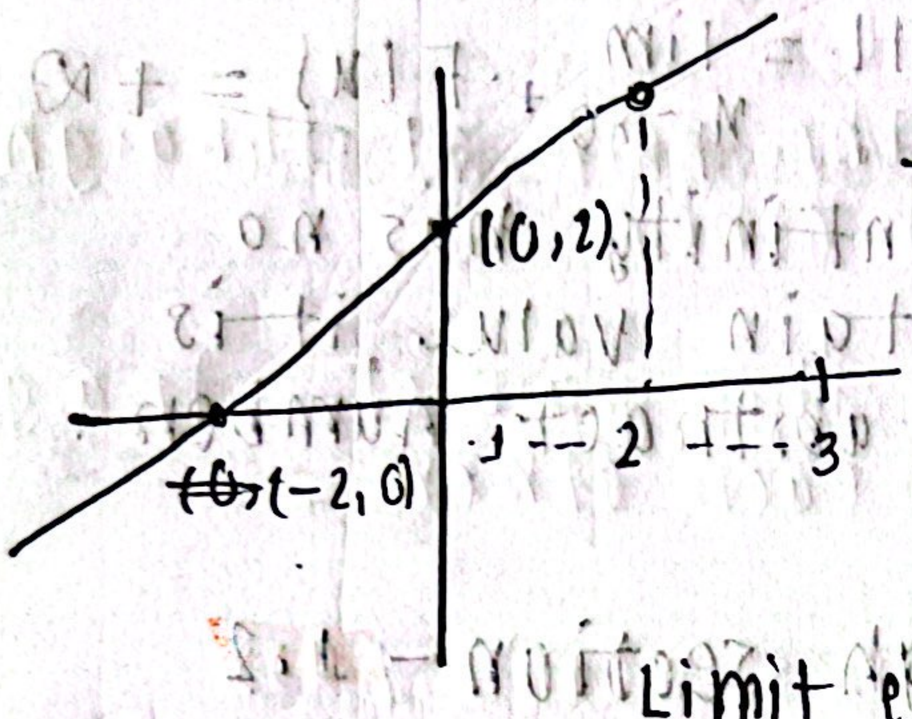
$f(x) = \frac{x^2 - 4}{x - 2}$; $R \setminus \{2\}$

$= \frac{(x+2)(x-2)}{(x-2)}$ → when we discard a part we can only discand it from the function only when, $x - 2 \neq 0$

$\Rightarrow x \neq 2$

$= (x+2)$; $x \neq 2$

Even we simplify the function, the simplified function will be still discontinuous at 2.



→ This graph is discontinuous yet limit exists at $x=2$.

$$LHL = \lim_{x \to 2^-} f(x) = 4$$

$$RHL = \lim_{x \to 2^+} f(x) = 4.$$

Limit exists but not continuous.

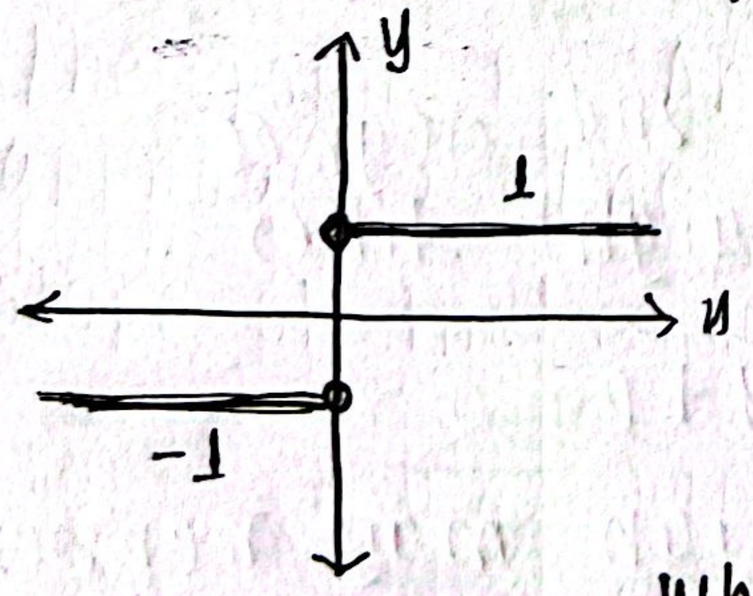
Section 1.1.1 Limits (an informal view) - Book

Example 3. Use numerical evidence to make a conjecture about the value of

$$\lim_{x \to 0} \frac{\sin x}{x}$$

One sided limits:

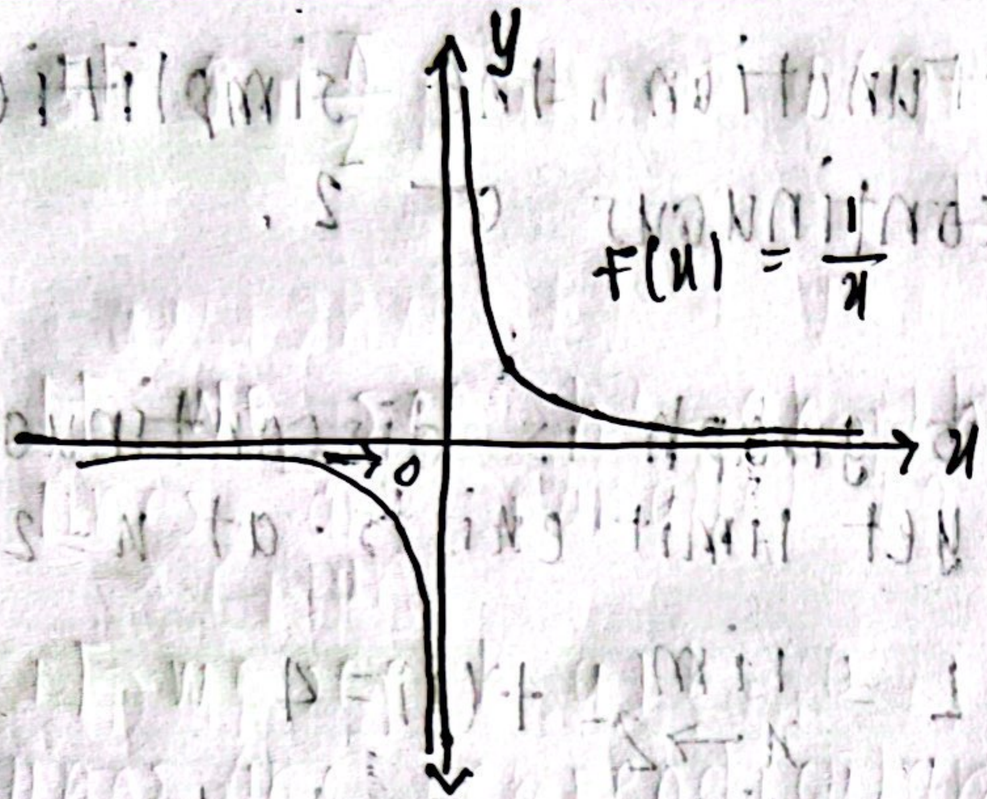
$$f(x) = \frac{|x|}{x} \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases} \rightarrow \text{Piece-wise Function.}$$



$$LHL = \lim_{x \to 0^-} f(x) = -1.$$

$$RHL = \lim_{x \to 0^+} f(x) = 1.$$

Limit will not exist as a whole but one sided limit exist.



LHL $= \lim_{x \to 0^-} F(x) = -\infty$

RHL $= \lim_{x \to 0^+} F(x) = +\infty$.

Infinity has no certain value it is an abstract number.

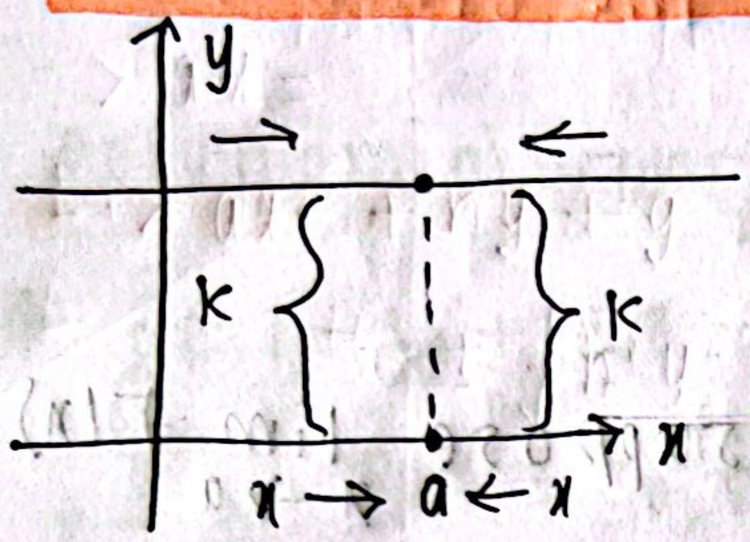
Home work - Read through section - 1.2 (all pages).

Exercise set 1.2 problems 3-30.

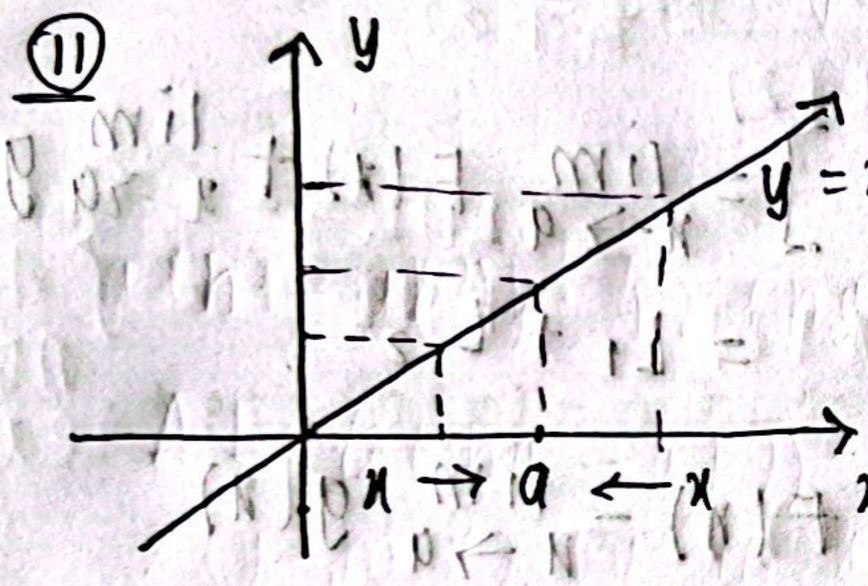
20/08/2024

Math 4141

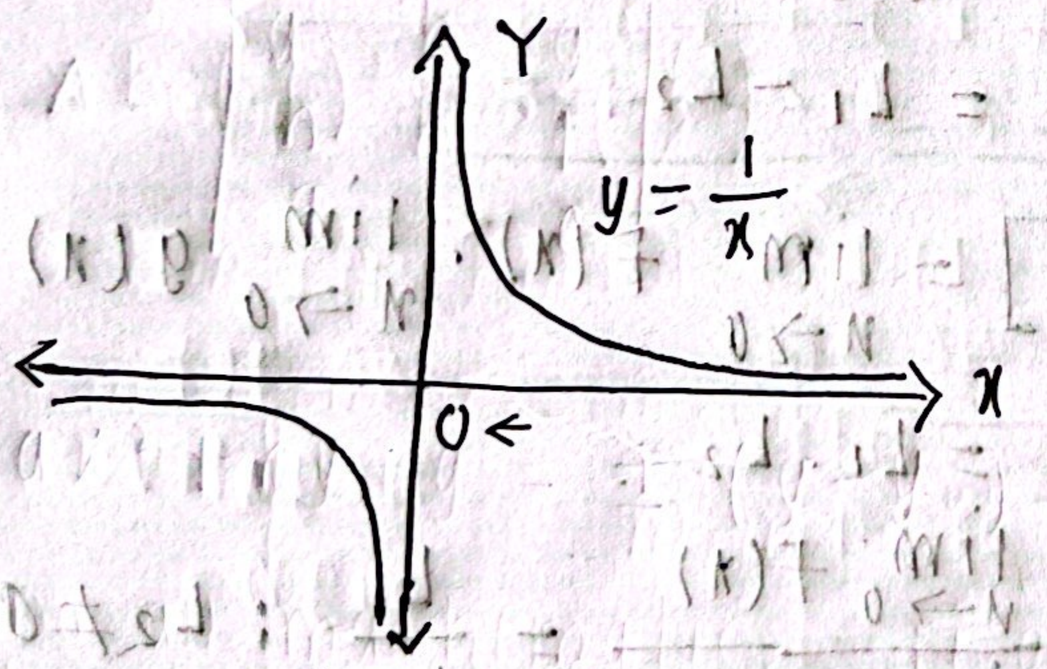
section 1.2 : some basic limits :



$y = k$ $\quad \lim_{x \to a} F(x) = \lim_{x \to a} k = k$ ①



② $\lim_{x \to a} F(x) = \lim_{x \to a} x = a$



③ $\lim_{x \to 0+} F(x) = \lim_{x \to 0+} \frac{1}{x} = +\infty$

$\lim_{x \to 0-} F(x) = \lim_{x \to 0-} \frac{1}{x} = -\infty$

Example: $F(a) = k$

$\lim_{x \to 3} 3 = 3$

$\lim_{x \to 5} 3 = 3$

→ For constant values the limit is equal to the constant it doesn't depend on the approaching value.

Example: $f(x) = x$

$\lim_{x \to 0} x = 0$

$\lim_{x \to 5} x = 5$

$\lim_{x \to 25} x = 25$

Theorem 1.2.2: Let $a \in \mathbb{R}$ & suppose $\lim_{x \to a} f(x) = L_1$, & $\lim_{x \to a} g(x) = L_2$.

Then, (i) $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$

$= L_1 + L_2$.

(ii) $\lim_{x \to a} [f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$

$= L_1 - L_2$.

(iii) $\lim_{x \to a} [f(x) \cdot g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$

$= L_1 \cdot L_2$.

(iv) $\lim_{x \to a} \left[\frac{f(x)}{g(x)}\right] = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} = \frac{L_1}{L_2}$; $L_2 \neq 0$.

(v) $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)} = \sqrt[n]{L_1} = L_1^{\frac{1}{n}}$;

$L_1 > 0$ if n is even.

Ex: $\lim_{x\to 5}(x^2-4x+3) = \lim_{x\to 5} x^2 - \lim_{x\to 5} 4x + \lim_{x\to 5} 3$

$= 25 - 20 + 3$

$= 8.$

polynomial function:

* $p(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$

$\lim_{x\to a} p(x) = c_0 + c_1a + c_2a^2 + \cdots + c_na^n$

$= p(a)$

Ex: $\lim_{x\to 1}(x^7 - 2x^5 + 1)^{\frac{3}{5}} = 0$

Let, $p(x) = x^7 - 2x^5 + 1$

$p(1) = 0$

Ex: $\lim_{x\to 2} \frac{5x^3+4}{x-3} = \frac{\lim_{x\to 2}(5x^3+4)}{\lim_{x\to 2}(x-3)}$

$= \frac{5\cdot 2^3 + 4}{2-3}$

$= -44.$

conditions when a national function ~~whe~~ have no limit:

Limits of rational function does not exist if one of the following situation occurs:

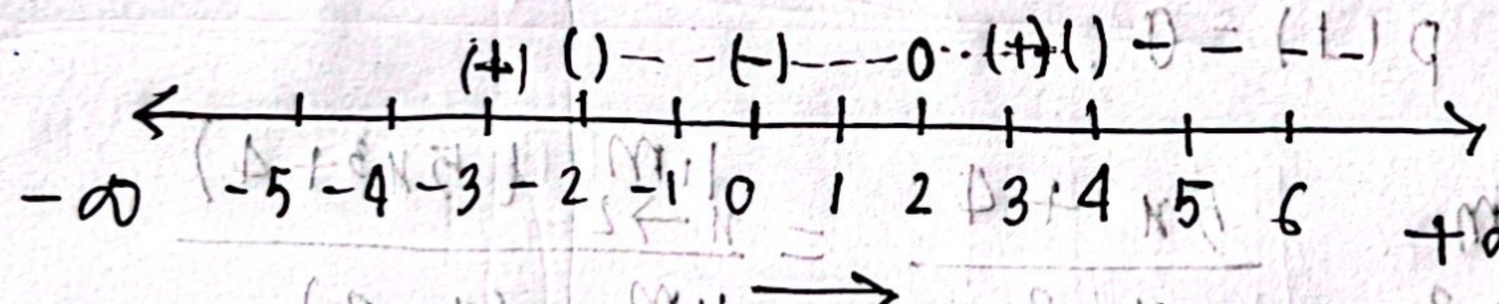
(i) The limit may be $-\infty$ from one side & $+\infty$ " the other side.

Examples: Evaluate limit

(a) $\lim_{x \to 4+} \frac{2-x}{(x-4)(x+2)} = \pm\infty$

(b) $\lim_{x \to 4-} \frac{2-x}{(x-4)(x+2)} = +\infty$

(c) $\lim_{x \to 4} \frac{2-(x+3)}{(x-4)(x+2)} =$ LHL $= +\infty$, RHL $= -\infty$ } Limit does not exist.

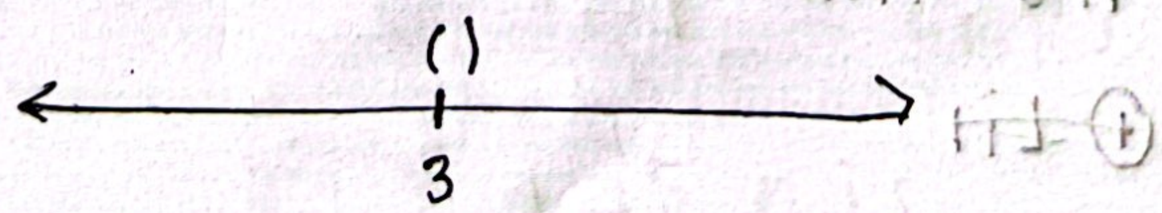


Ex:

a) $\lim_{x \to 3} \frac{x^2-6x+9}{x-3}$

$= \frac{2x-6}{1}$

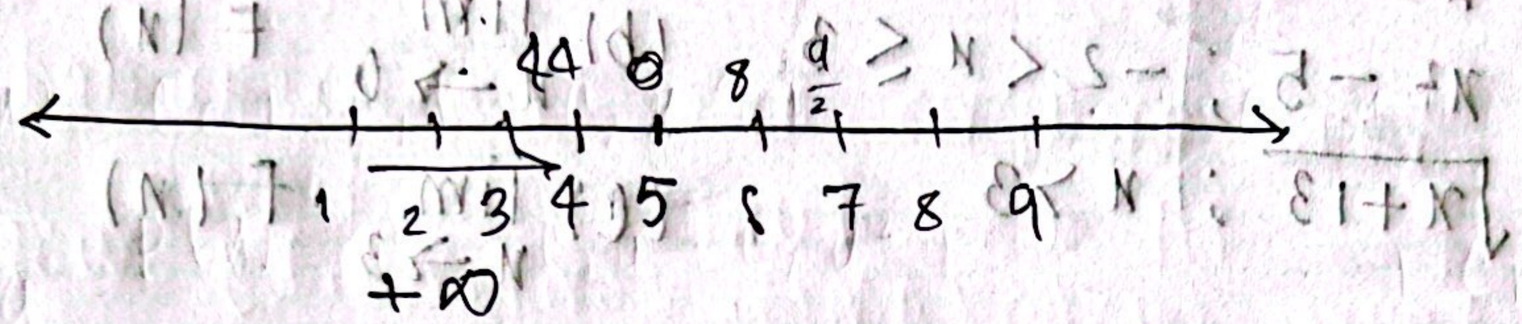
$= \frac{6-6}{1} = 0$



$$\frac{2(x+4)}{(x-3)(x+4)} = \frac{2}{x-3} \quad \therefore \frac{2}{-4-3} = \frac{-2}{7}$$

(b) $\lim_{x \to -4} \frac{2x+8}{x^2+x-12} = \lim_{x \to -4} \frac{2}{2x+1} = \frac{2}{-8+1} = \frac{-2}{7}$

(c) $\lim_{x \to 5} \frac{x^2-3x-10}{x^2+10x+25} = \lim_{x \to 5} \frac{(x-5)(x+2)}{(x-5)^2} = \lim_{x \to 5} \frac{x+2}{x-5}$



Theonem: (1.2.4)

Let $F(x) = \frac{P(x)}{q(x)}$ be a national Function.

Let, $a \in R$.

(i) IF $q(a) \neq 0$, then $\lim_{x \to a} F(x) = F(a)$.

(ii) IF $q(a) = 0$, but $p(a) \neq 0$, then

$\lim_{x \to a} F(x)$ does not exist.

Finding limits of radicals.

Ex: Find $\lim_{x \to 1} \frac{x-1}{\sqrt{x}-1}$

$= \lim_{x \to 1} (\sqrt{x}+1) = 2$.

Piecewise-Defined Functions (Limit)

sketching of graph needed for solving problems like these

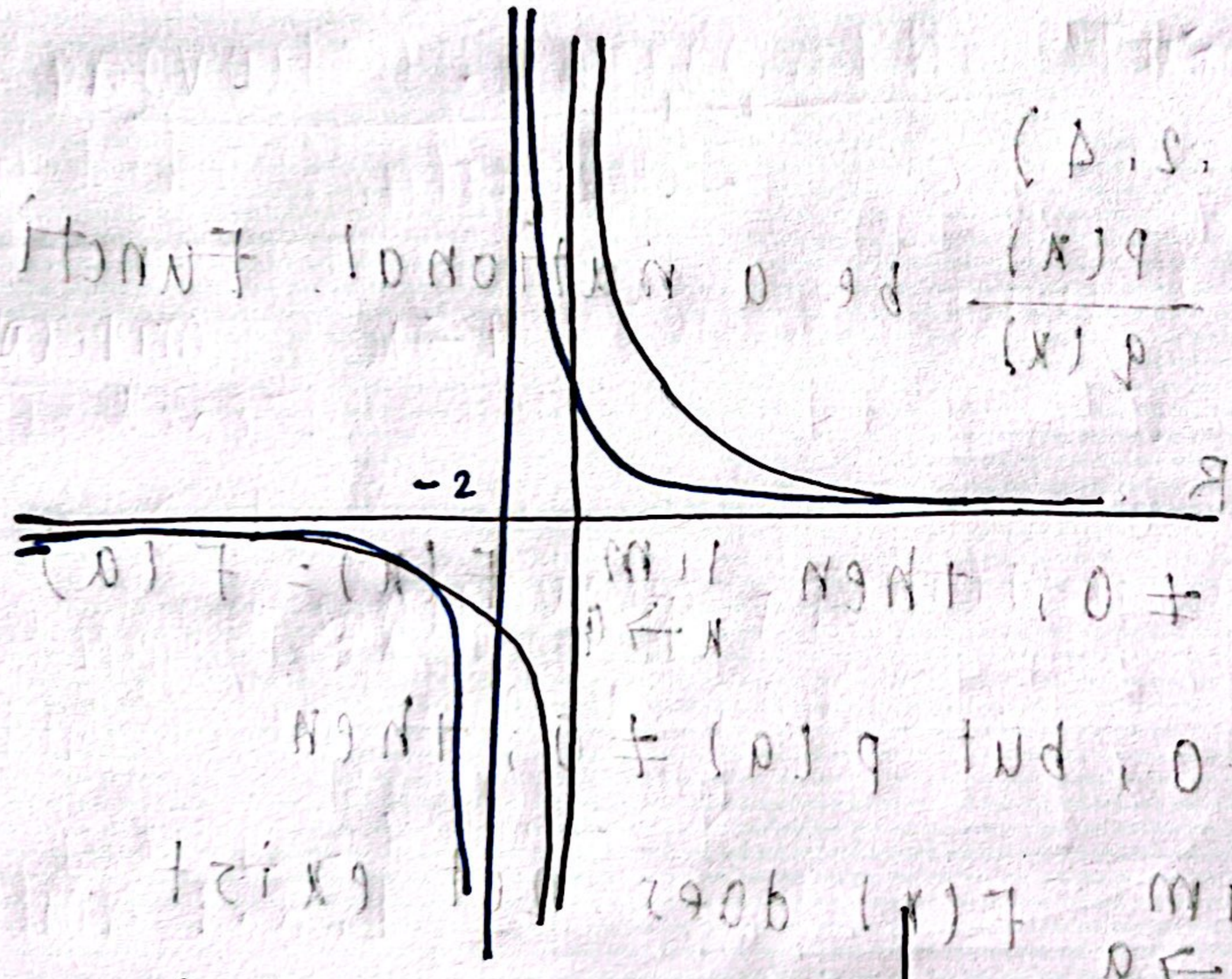
Ex:

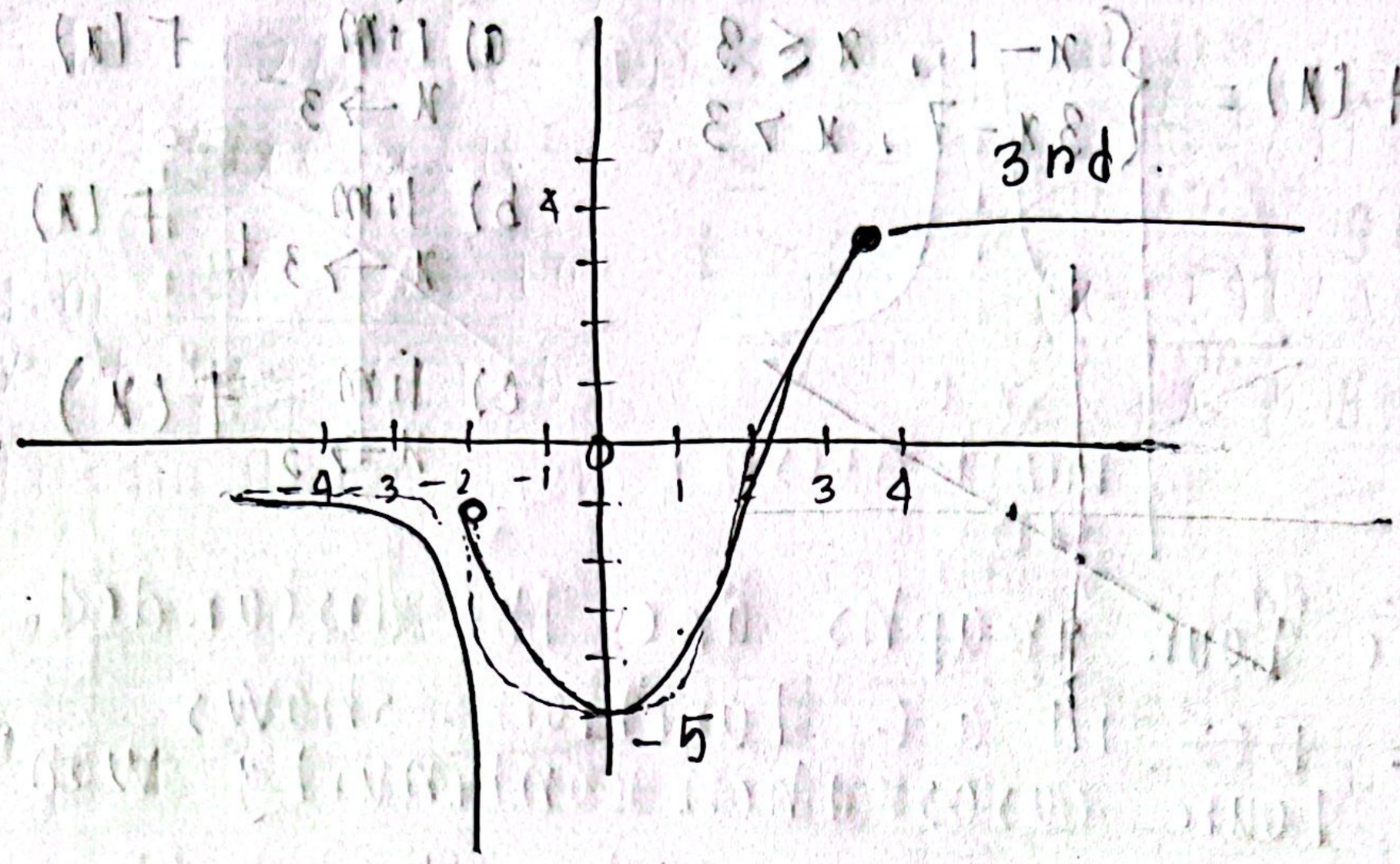
$$f(x)=\begin{cases}\dfrac{-1}{x+2}; & x<-2\\ x^2-5; & -2<x\le 3\\ \sqrt{x+13}; & x>3\end{cases}$$

Find (a) $\lim_{x\to -2} f(x)$

(b) $\lim_{x\to 0} f(x)$

(c) $\lim_{x\to 3} f(x)$





(a) $\lim_{x \to -2} F(x)$ LHL $= +\infty$, RHL $= -1$. Limit doesn't exist

(b) $\lim_{x \to 0} x^2 - 5 =$ LHL $= -5$, RHL $= -5$. Limit does exist.

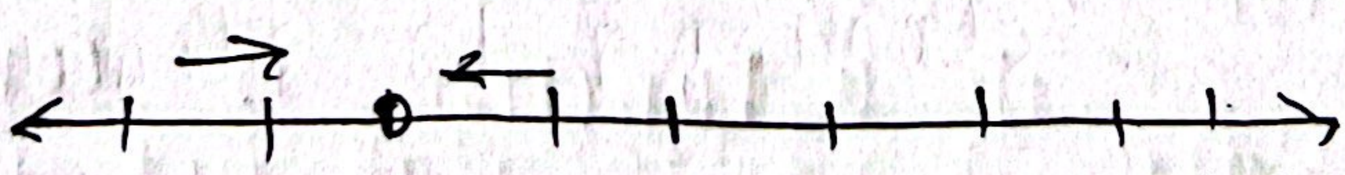
(c) $\lim_{x \to 3} F(x) = \lim_{x \to 3}$

LHL $= 4$ (2nd)

RHL $= 4$ (3rd)

LHL = RHL

Limit does exist.

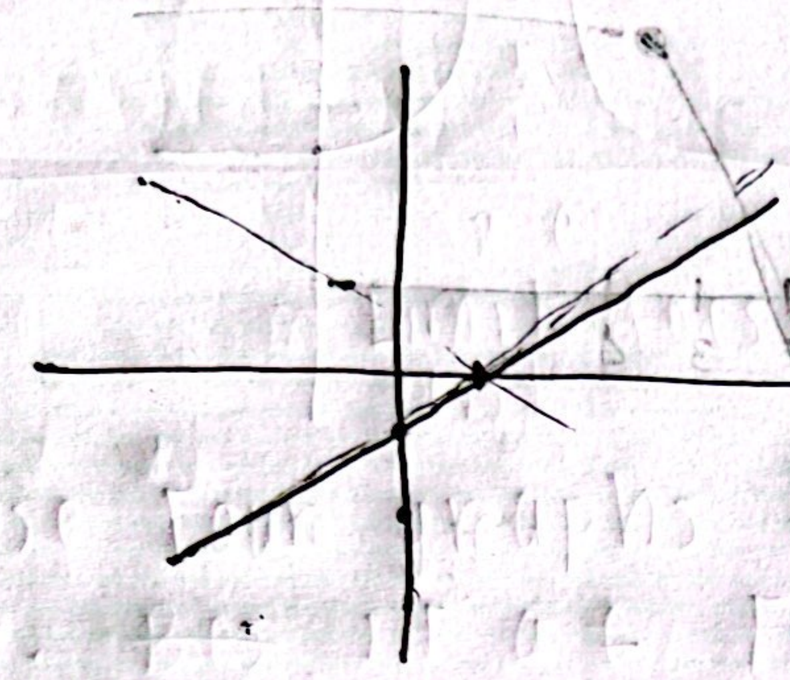


$-\infty$ -4 -3 -2 -1 0 1 2 3 4 $+\infty$

$x = 1$

$x = \frac{3}{7}$

31) $F(x) = \begin{cases} x-1, & x \leq 3 \\ 3x-7, & x > 3 \end{cases}$



a) $\lim_{x \to 3^-} F(x)$

b) $\lim_{x \to 3^+} F(x)$

c) $\lim_{x \to 3} F(x)$ ...

a) $\lim_{x \to 3^-} x - 1$

$= 3 - 1 = 2.$

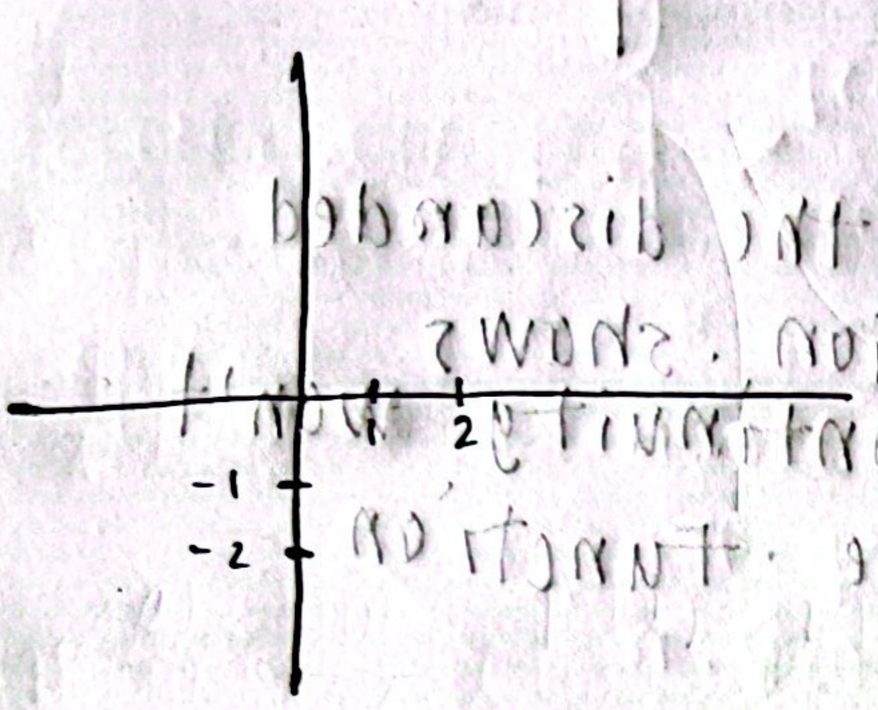
b) $\lim_{x \to 3^+} 3x - 7 = 9 - 7 = 2$

L.H.L = R.H.L = 2

c) $\lim_{x \to 3} x - 1 = 3 - 1 = 2.$

32) Let,

$$g(t) = \begin{cases} t-2, & t < 0 \\ t^2, & 0 \le t \le 2 \\ 2t, & t > 0 \end{cases}$$



(a) $\lim_{t \to 0} g(t)$

(b) $\lim_{t \to 1} g(t)$

(c) $\lim_{t \to 2} g(t)$ → L.H.L = −2

(a) $\lim_{t \to 0} g(t) = \lim_{t \to 0} t^2$

$= 0$ → R.H.L =

limit does not exist

(b) $\lim_{t \to 1^-} t = 1$

$\lim_{t \to 1^+} t^2 = 1$

Limit exist

(c) $\lim_{t \to 2} g(t)$ = LHL $= \lim_{t \to 2^-} g(t) = \lim_{t \to 2^-} t^2$

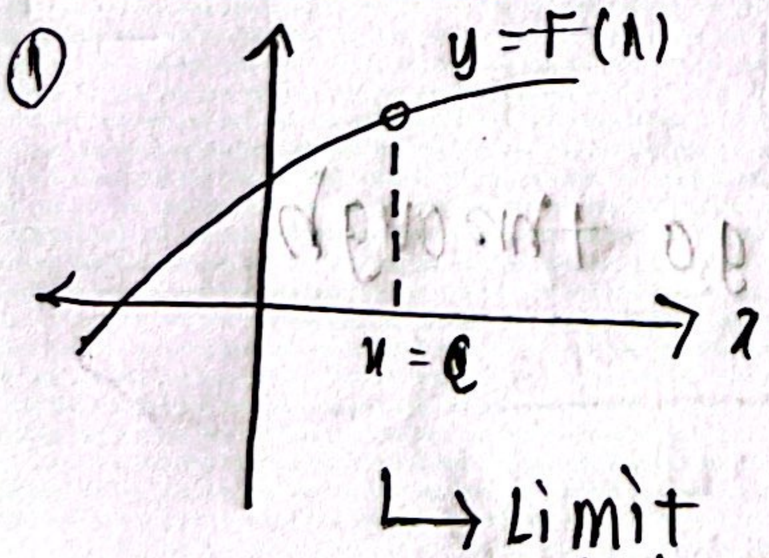
$= 4$

R.H.L $= \lim_{t \to 2^+} g(t) = \lim_{t \to 2^+} 2t = 2 \cdot 2 = 4$

L.H.L = R.H.L = 4

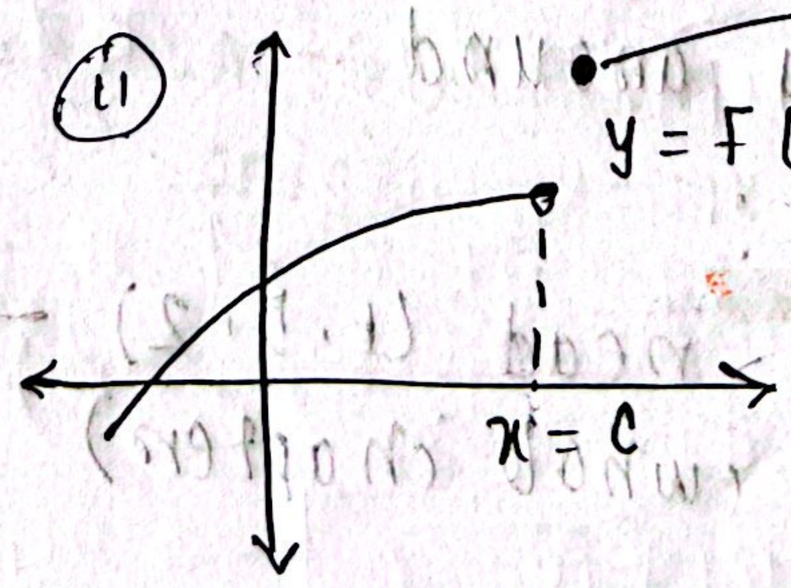
continuity :

(i)



→ Limit exist

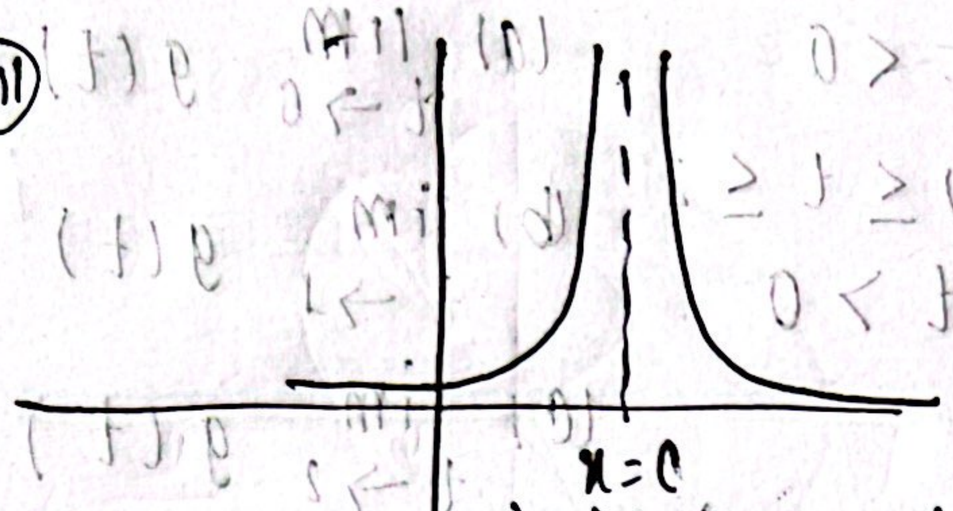
(ii)



→ height difference RHL ≠ LHL limit doesn't exist discontinuous

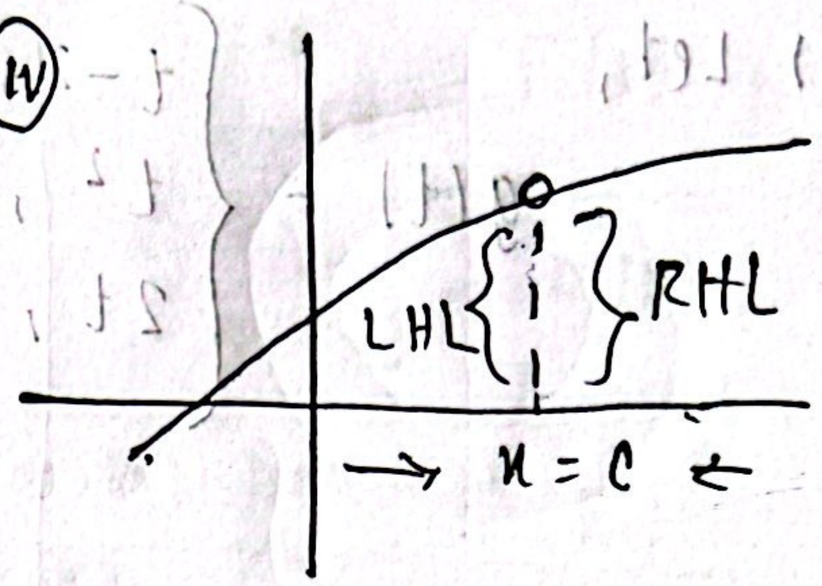
LHL ⟶ ⟵ RHL

(iii)



Limit doesn't exist.

(iv)



These four graphs are the discarded cases. ~~Bes~~ if a function shows these four cases then continuity won't be prevailed. Hence, the function won't be continuous.

(iv) Limit exist but not equal to the functional value

* To be a continuous, function needs to be defined in every points.

Section 1.5.1

If a function is differential it must be continuous & if a function is continuous then limit exist. But not the other way around.

Homework:

Definition - read (1.5.2) → go through (whole chapter)

Limit & continuity

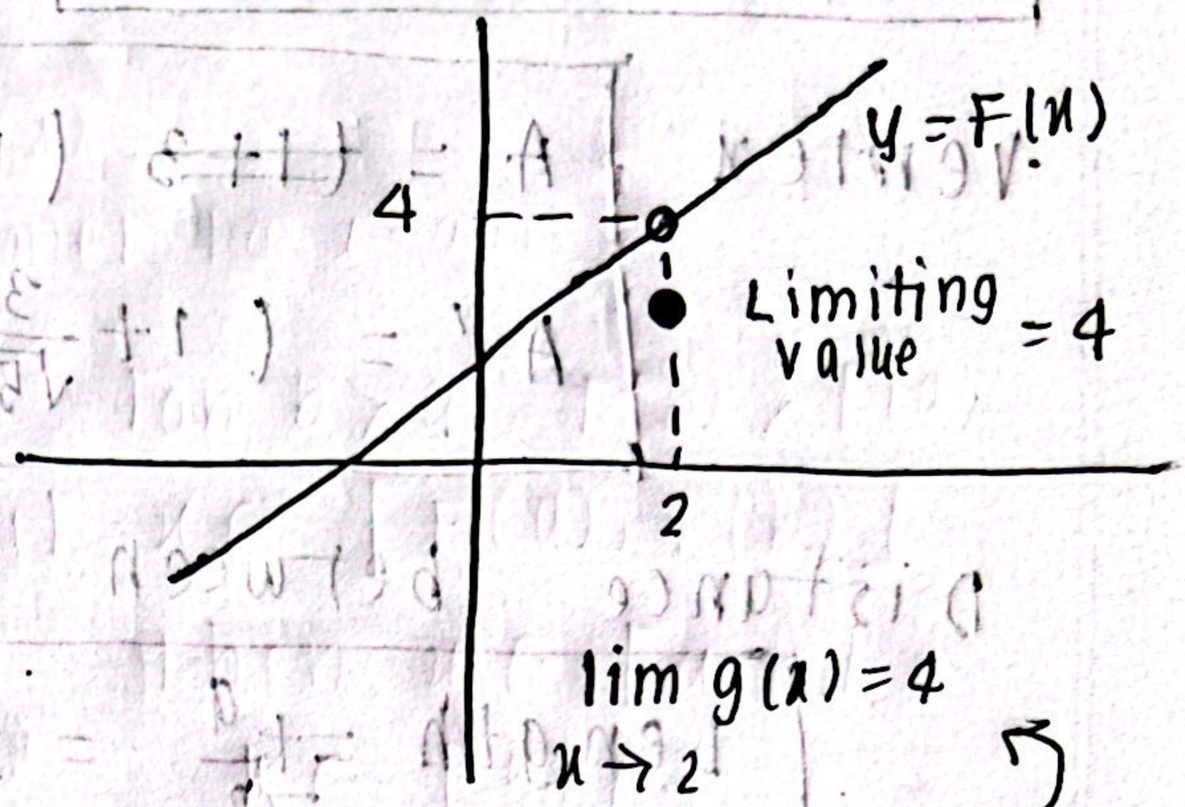
27/09/2024

Example:

$$F(x) = \frac{x^2-4}{x-2}$$

$$g(x) = \begin{cases} \frac{x^2-4}{x-2} ; & x \neq 2 \\ 3 ; & x = 2 \end{cases}$$

$$h(x) = \begin{cases} \frac{x^2-4}{x-2} ; & x \neq 2 \\ 4 ; & x = 2 \end{cases}$$



- What even limit I take

$g(2) = 3$

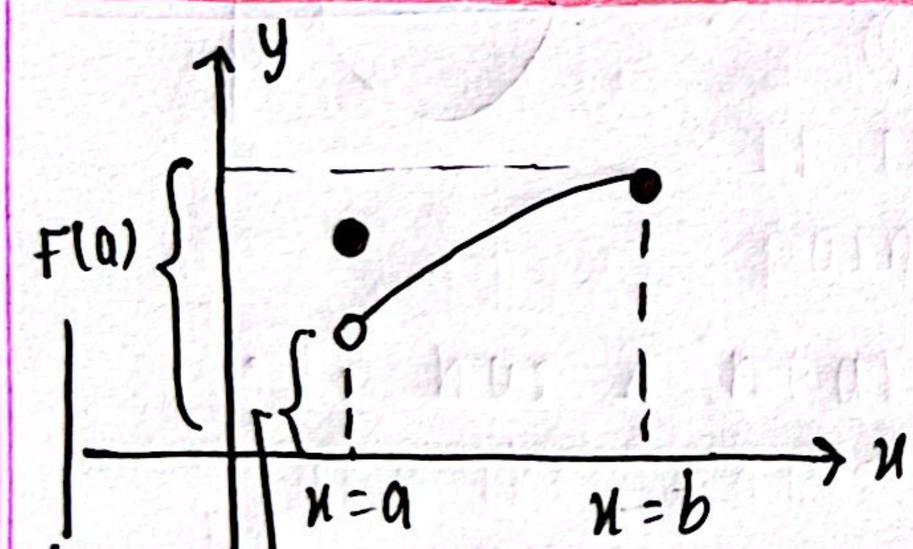
∴ Limiting Value ≠ Function Value.

So, ~~clìston~~ discontinuous.

- For $h(x)$ =

Limiting Value = Function Value.

continuity on an Interval: $[a, b]$



height defines the function value.

move in this point to reach the right end point.

x approaches b.

limiting value height.

continuous at the right end point $[a, b]$.

~~Not co~~. if $\lim_{x \to b^-} F(x) = F(b)$.

Not continuous, at the left end point on $[a, b]$.

if $\lim_{x \to a^+} F(x) \neq F(a)$.

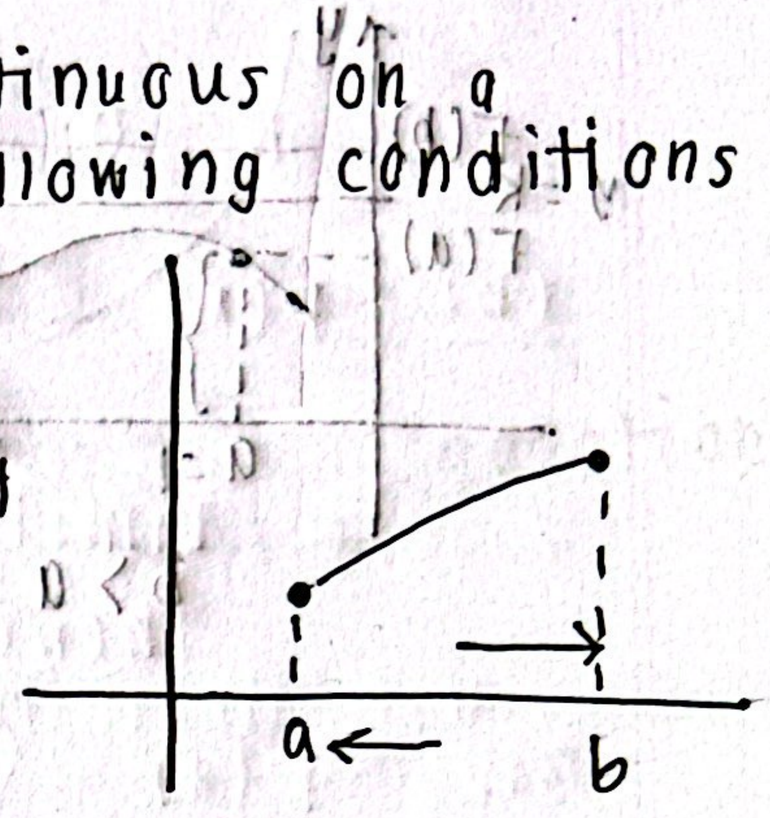
Defⁿ : (1.5.2)

A function f is said to be continuous on a closed interval $[a, b]$ if the following conditions are satisfied :

(i) f is continuous on (a, b)

(ii) f is continuous from the right at a. $\lim_{x \to a^+} F(x) = F(a)$

(iii) f is continuous from the left at b. $\lim_{x \to b^-} F(x) = F(b)$



N.B : • To reach the left end the f is continuous from the right at a.

$\lim_{x \to a^+} F(x) = F(a)$.

• To reach the right end the f is continuous from the left at b.

$\lim_{x \to b^-} F(x) = F(b)$

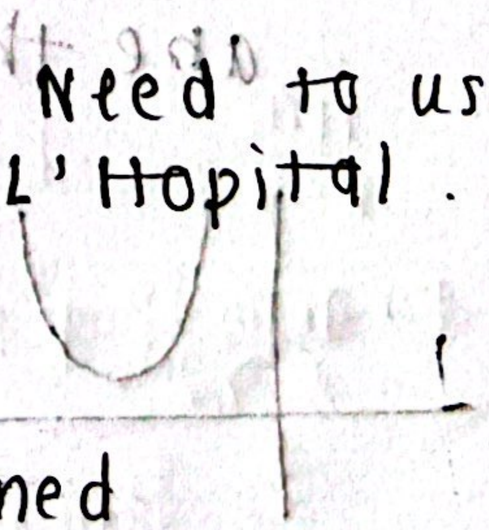
* A polynomial is continuous from all points.

Example: $y = F(x) = \dfrac{x^2 - 9}{x^2 - 5x + 6}$ → Need to use L'Hopital.

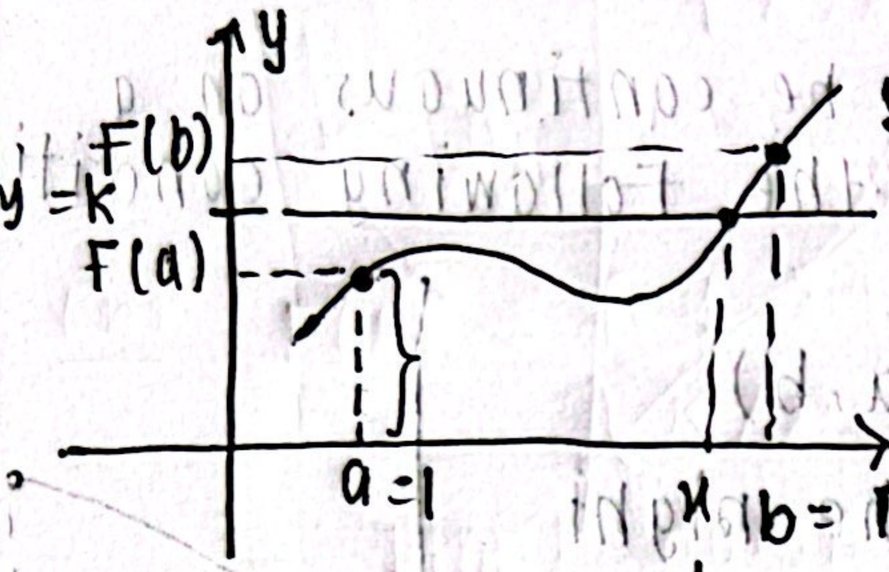
$x^2 - 5x + 6 = 0$

$\Rightarrow (x-2)(x-3) \neq 0$

$\therefore x \neq 2, 3$ → Not defined



1.5.7 The intermediate-value theorem :



parallel to x-axis

For $b > a$, $F(b) > F(a)$

if $u \in [F(a), F(b)]$

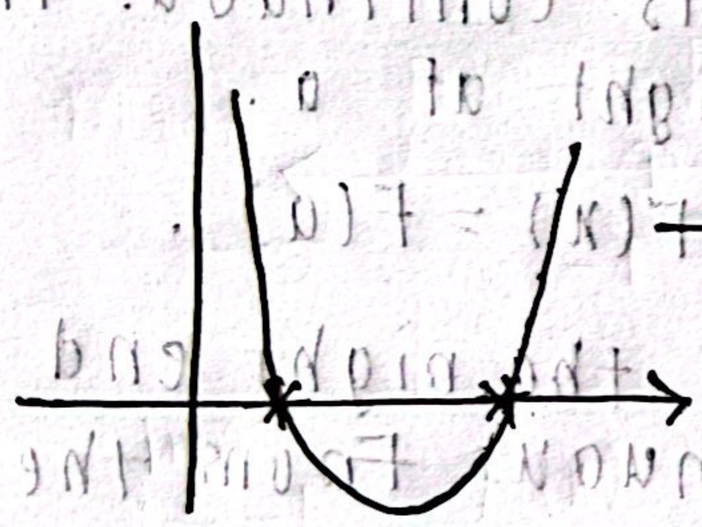
then $u \in [a, b]$

$b > a$

For this u value it is cutting the curve.

$k \in [5, 7]$

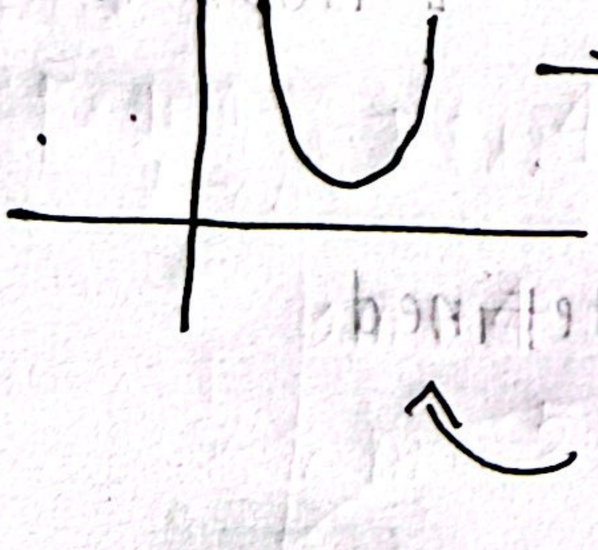
$u \in [1, 10]$



→ This is a curve of two degree polynomial function
Quadratic equation.

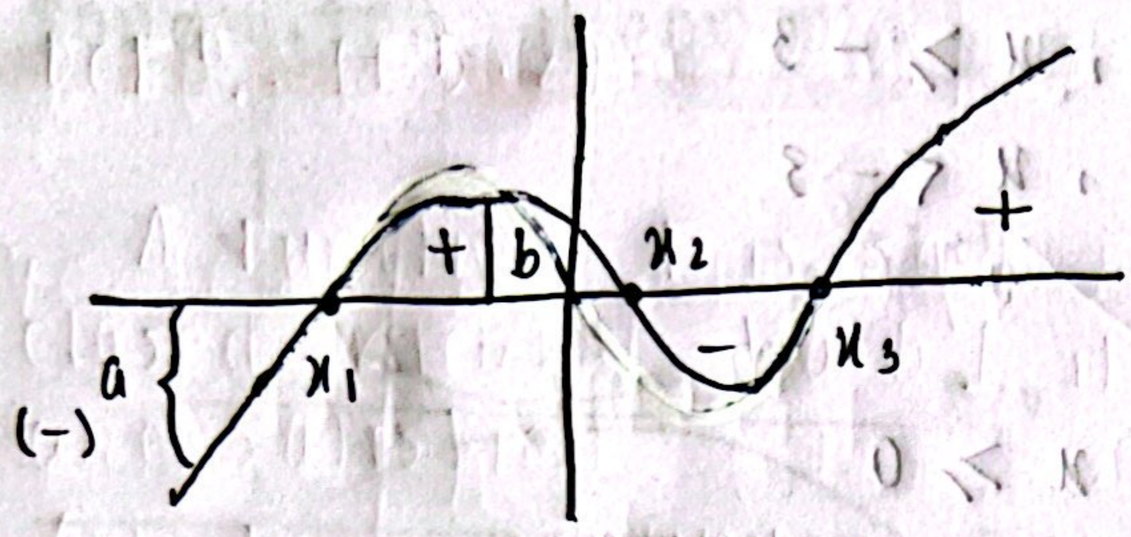
It cuts the x-axis in two points, so, 2-degree polynomial.

- It has 2 real roots. i.e. the points are the roots.



→ Doesn't cut x-axis in any points. so, roots are not real.

Imaginary roots.



Home work - 29, 30, 31, 36 :- Recommended questions From ~~15~~ : 1.5.2

29.(a)

$$F(x) = \begin{cases} 7x-2 \; ; & x \leq 1 \\ kx^2 \; ; & x > 1 \end{cases}$$

$(-\infty, 1)$

$-\infty \longleftarrow 1 \longrightarrow +\infty$ → subdomains need to find the values from these subdomains.

solution: F is continuous for $x < 1$ & for $x > 1$.

$$L.H.L = \lim_{x \to 1^-} F(x) = 5$$

$$R.H.S = \lim_{x \to 1^+} F(x) = k.$$

For continuous,

$L.H.S = R.H.S$

$k = 5$.

29.(b)

$$F(x) = \begin{cases} kx+1, & x \leq 2 \\ 2x+k, & x > 2 \end{cases}$$

$$L.H.L = \lim_{x \to 2^-} kx^2 = 4k.$$

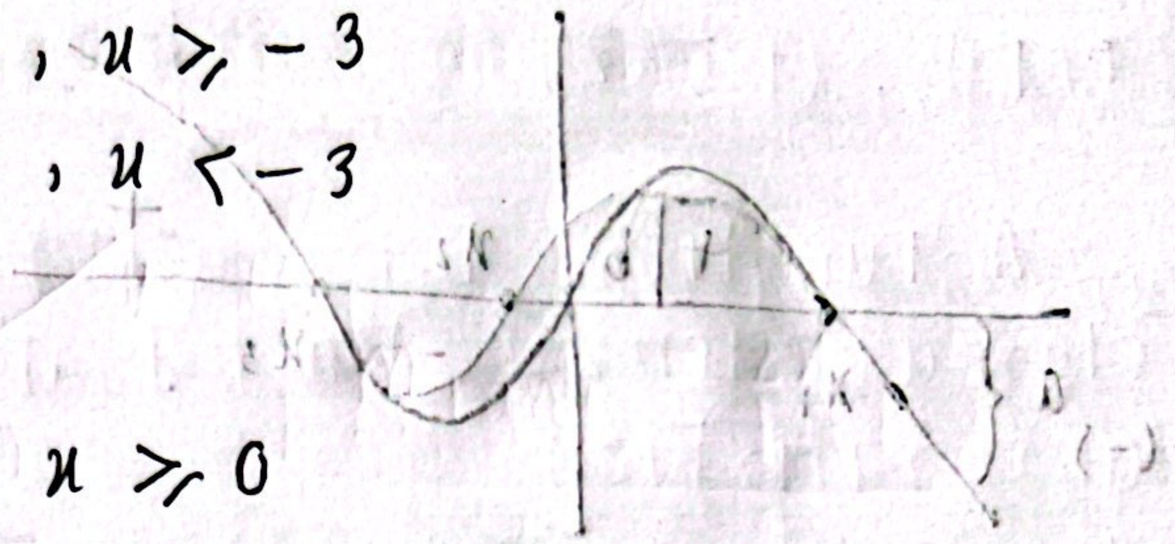
$$R.H.L = \lim_{x \to 2^+} 2x+k = k+4$$

$$4k = k+4.$$

$$\Rightarrow k = \frac{4}{3}.$$

30. (a) $$F(x) = \begin{cases} 9-x^2, & x \geq -3 \\ k/x^2, & x < -3 \end{cases}$$

$K = 0$.

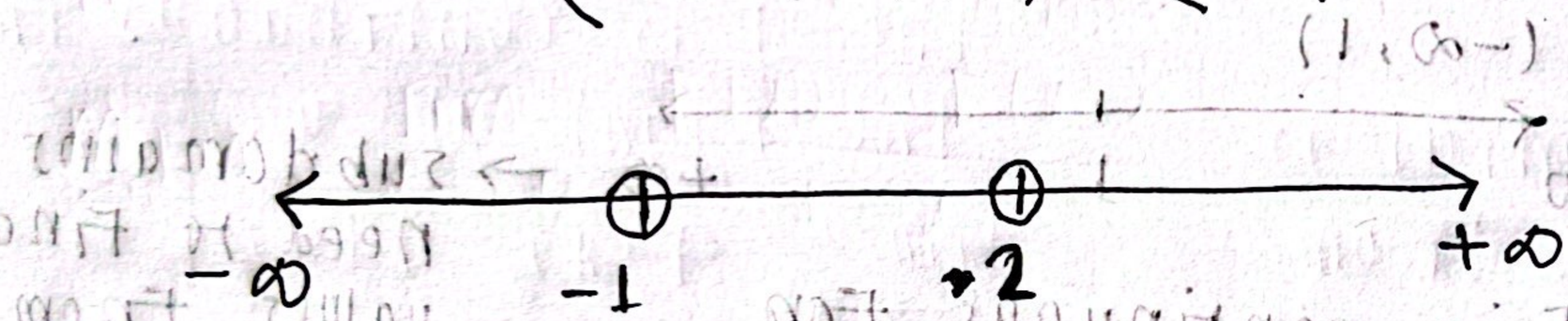


(b) $$F(x) = \begin{cases} 9-x^2, & x \geq 0 \\ k/x^2, & x < 0 \end{cases}$$

Limit doesn't exist ?

There is no value For K at 0.

31. $$F(x) = \begin{cases} x^2+5; & x > 2 \\ m(x+1)+k; & -1 < x \leq 2 \\ 2x^3+2+7; & x \leq -1 \end{cases}$$



$\lim_{x \to -1^-} F(x) = 4$

$\lim_{x \to +1^-} F(x) = k$

$\therefore K = 4$

$\lim_{x \to 2^-} F(x) = 3m+k = 3m+4$.

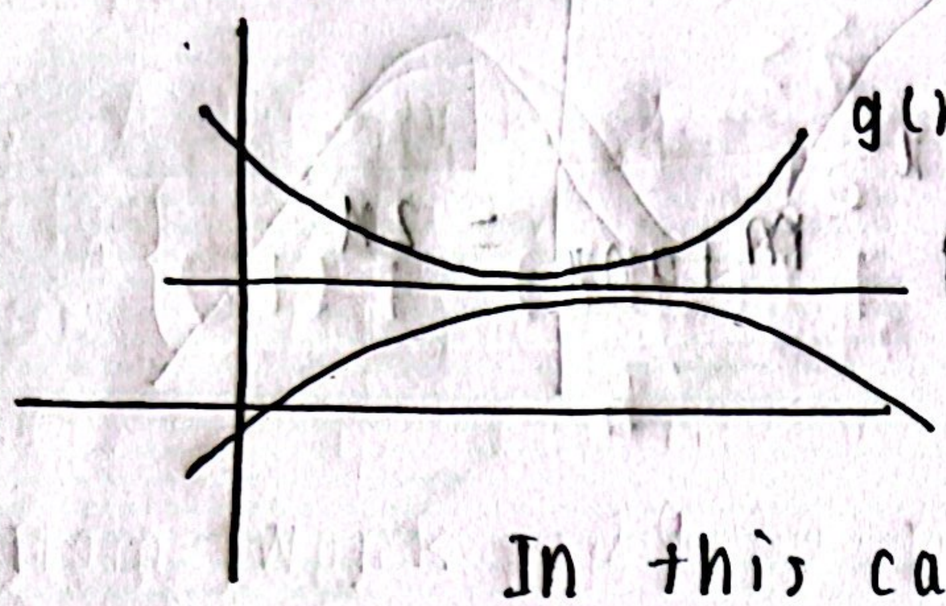
$\lim_{x \to 2^+} F(x) = 9$

$3m+4 = 9$

$\Rightarrow m = 5/3$.

H.W = 36

Sandwitch theonem :



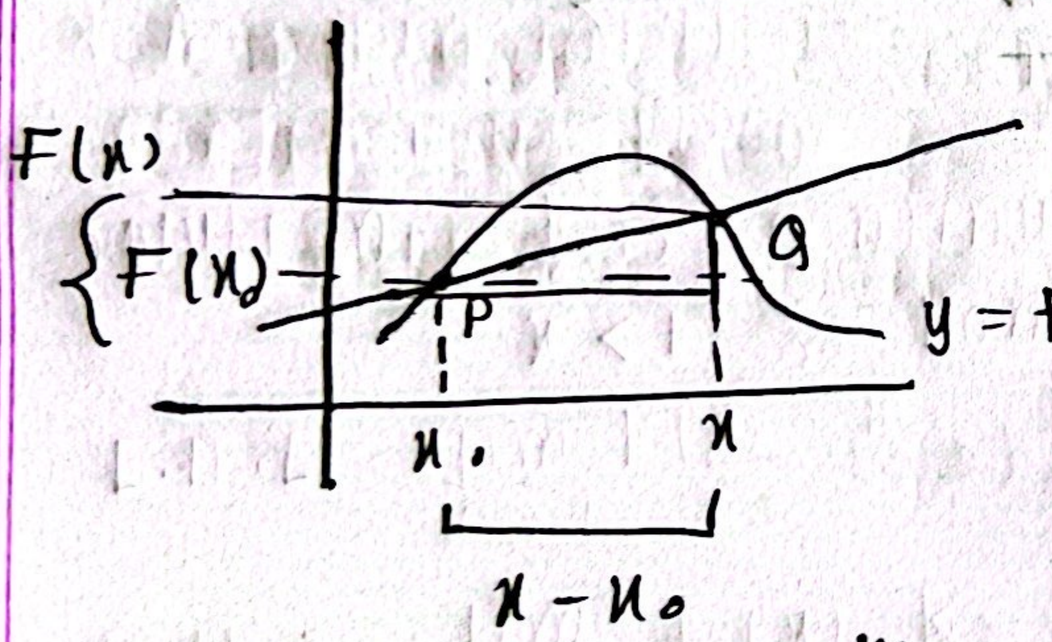
All these functions approaches the same point.

$F(x) \leq h(x) \leq g(x)$.

In this case we find the value of upper function & lower function.

That way we can find the value of middle value

2.1 : Derivative.



$P(x_0, F(x_0))$

$Q(x, F(x))$

$$m_{sec} = \frac{F(x) - F(x_0)}{x - x_0}$$

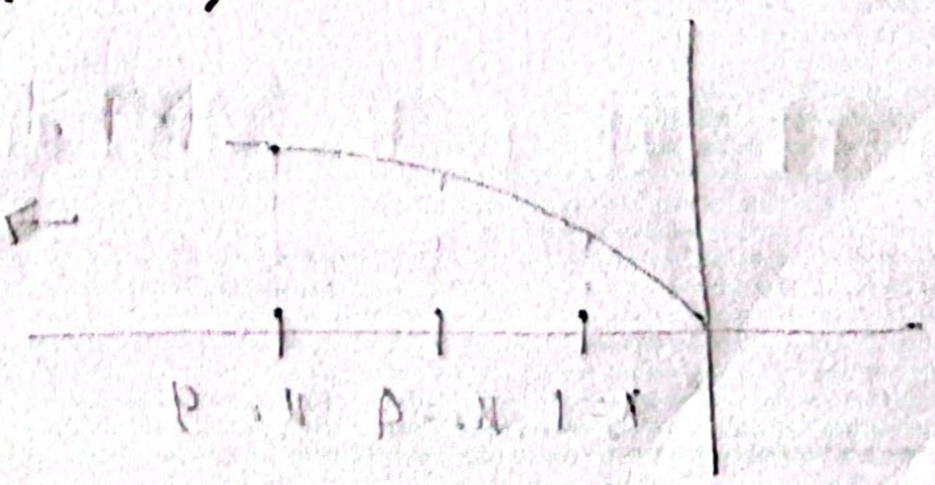
$$= \frac{F(x_0+h) - F(x_0)}{h}$$

$$m_{tan} = \lim_{x \to x_0} \frac{F(x) - F(x_0)}{x - x_0}$$

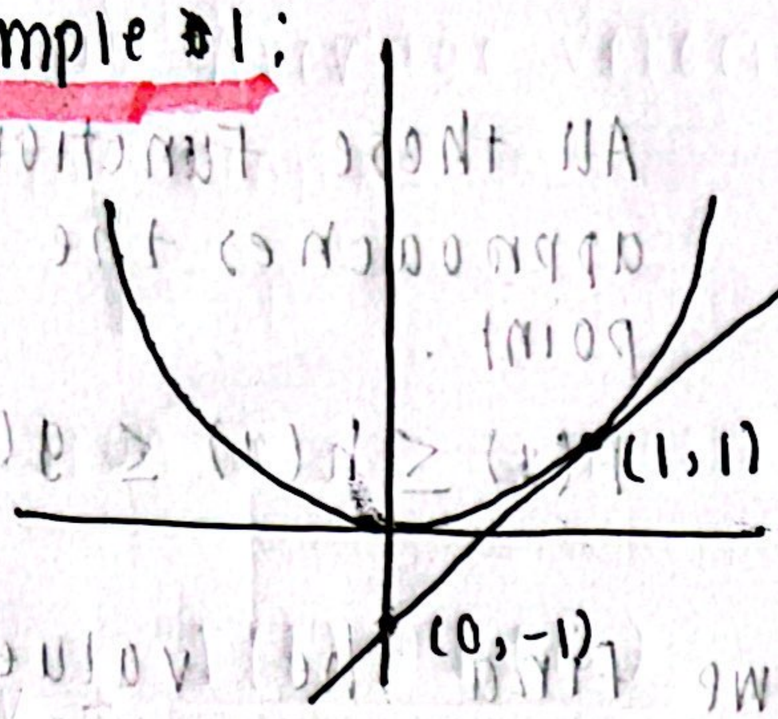
$$= \lim_{x \to x_0} \frac{F(x_0+h) - F(x_0)}{x - x_0}$$

$$y - y_0 = m(x - x_0)$$

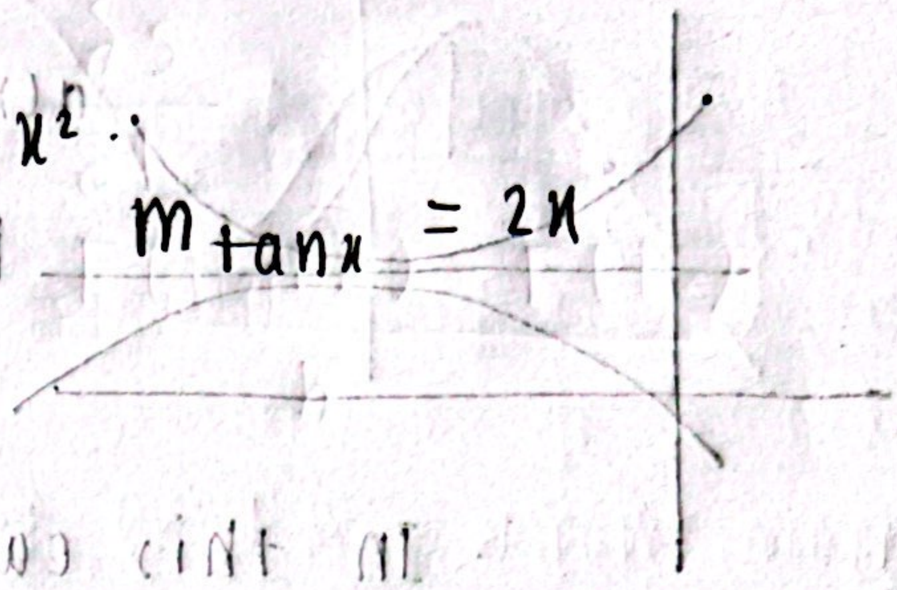
$$\Rightarrow F(x) - F(x_0) = m_{tan}(x - x_0)$$



Example #1;



$y = x^2$.



use definition 2.1 to find an equation for the tangent line to the parabola $y = x^2$ at the point (1,1), and confirm the result agrees with that obtained in Example 1 of section 1.2

$$m_{tan} = \lim_{x \to x_0} \frac{F(x_1) - F(x_0)}{x - x_0}$$

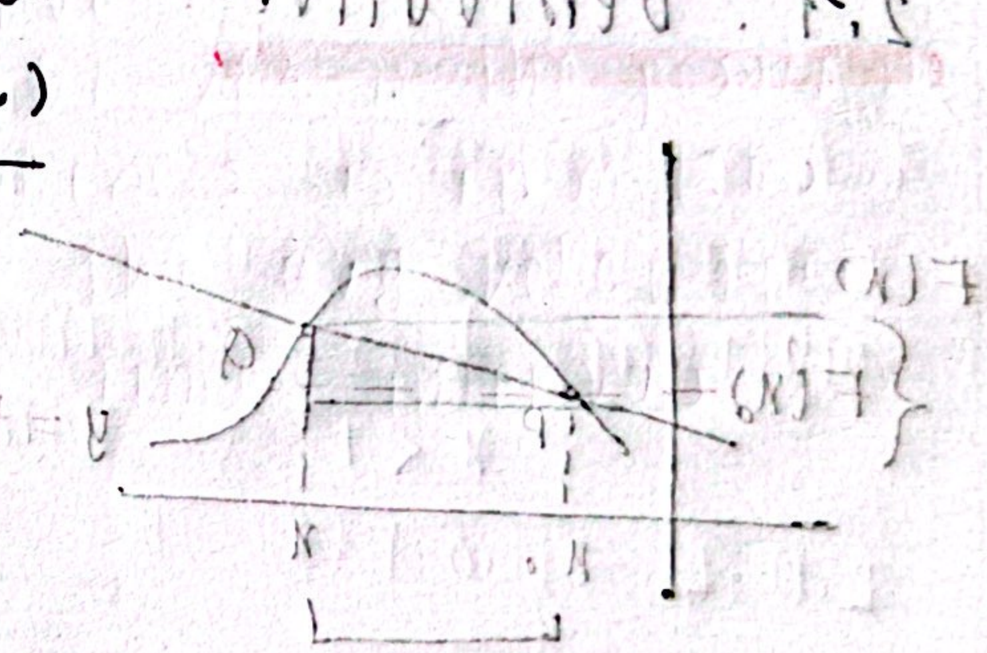
$$= \lim_{x \to 1} \frac{x^2 - 1}{x - 1}$$

$$= \lim_{x \to 1} (x+1) = 2.$$

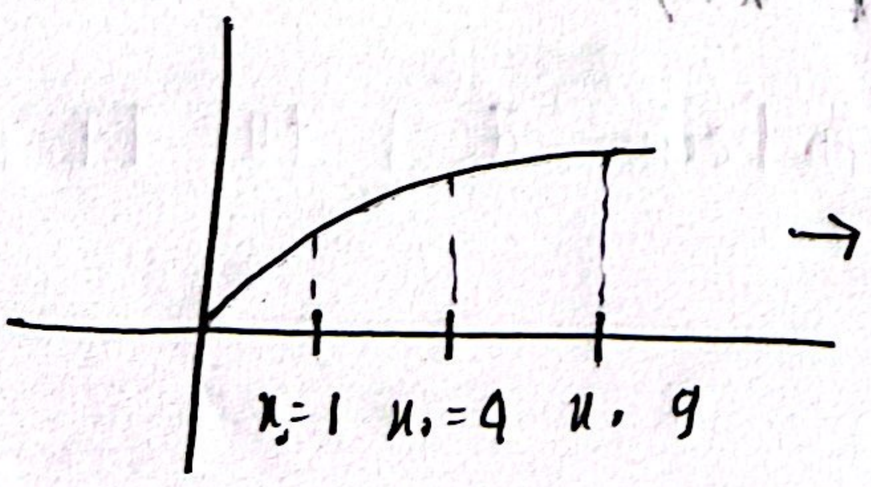
$$y - y_0 = m_{tan}(x - x_0)$$

$$\Rightarrow y - 1 = 2(x - 1)$$

$$\Rightarrow y = 2x - 1.$$



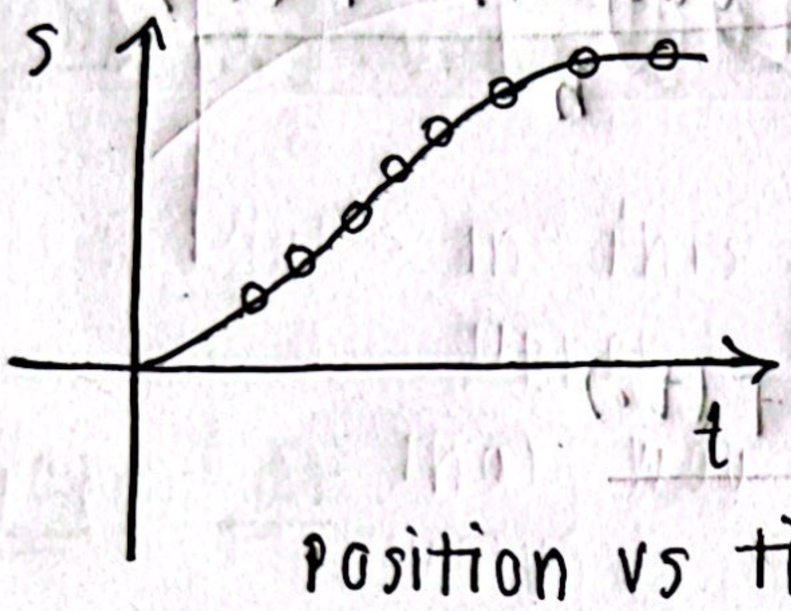
Example - 3, 4



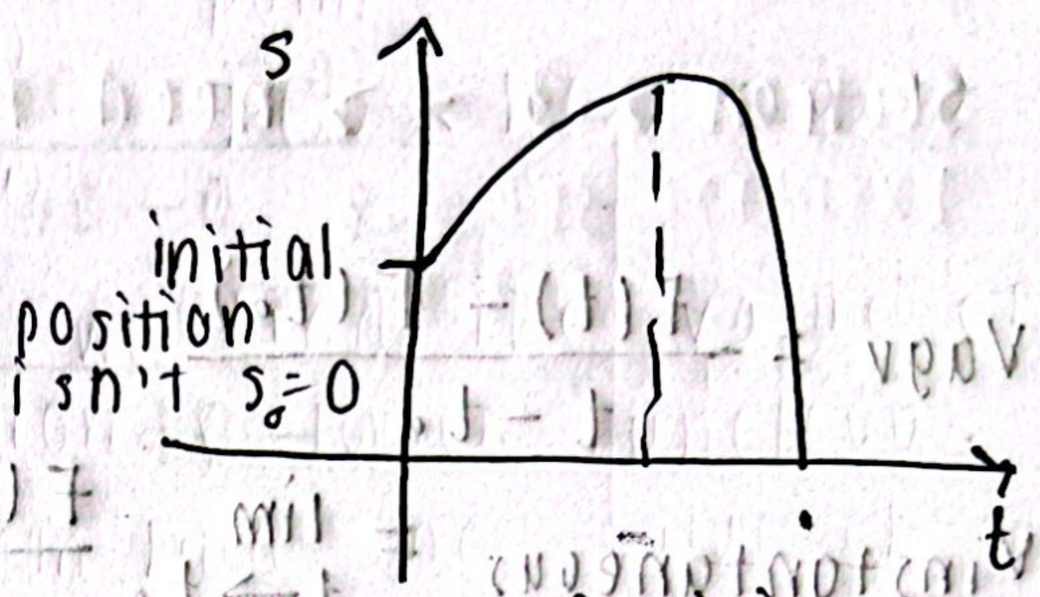
→ Example - 4.

Velocity:

IF object moves in one linear path it is called. rectilinear motion.



position vs time graph for rectilinear motion



Position vs time graph for projectile

After every differentiation, the degree of a function gets reduced by a degree.

For example, $\frac{d}{dx}(x^3) = 3x^2$; $\frac{d}{dx}(3x^2) = 6x$

$\frac{d}{dx}(6x) = 6$

projectile - when we analyze the motion it moves along a curve.

When we imagine a motion in one dimension then we can consider all motion in a rectilinear motion

↳ calculate position vs time.

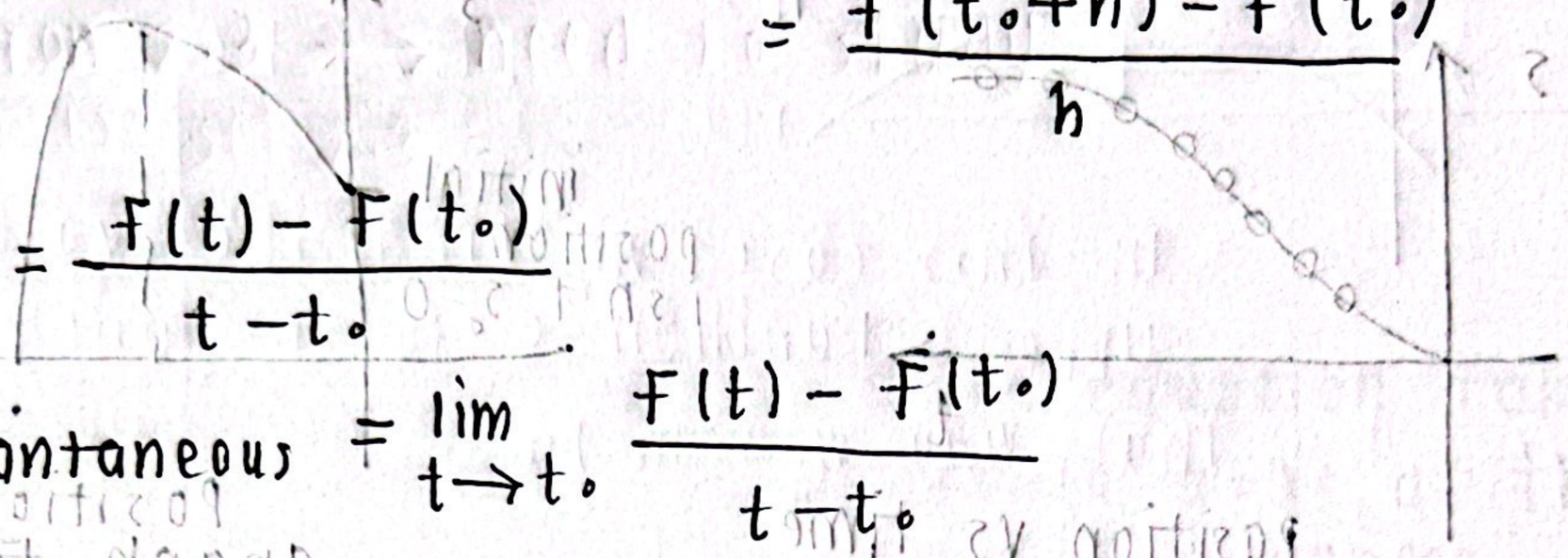
Rectilinear Motion: We need to consider only, $s = F(t)$ → displacement = function of tim

$$V_{avg} = \frac{\text{change in position}}{\text{time elapsed}} = \frac{F(t_0+h) - F(t_0)}{h}$$

~~Avega~~ Average velocity = m secant line

$$= \frac{\delta s}{\delta t}$$

$$= \frac{f(t_0+h) - f(t_0)}{h}$$



$$V_{avg} = \frac{f(t) - f(t_0)}{t - t_0}$$

$$V_{instantaneous} = \lim_{t \to t_0} \frac{f(t) - f(t_0)}{t - t_0}$$

Example - 5 ⇒ Rectilinear motion can trace the path of the curve

Suppose that $s = F(t) = 1 + 5t - 2t^2$ is the position function of a particle, where s is in meters & t is in seconds. Find the avg velocities of the particle over the time intervals (a) $[0, 2]$ & (b) $[2, 3]$

Example - 9

Exercise - 2.1 — 26, 27 ⇒ Recommended.

Exercise (26)

$s = F(t) = 1250 - 16t^2$ → position function with respect to time.

(a) If at $t = 5$, distance is not zero

Example 27 :

$S = 0.3t^3 \longrightarrow$ position Function w.r.t time.

a) $S = 0.3(40)^3$

b) Avg. velocity $= \dfrac{F(40) + F(0)}{40 - 0}$

$= \dfrac{0.3(40)^3 - F(0)}{40 - 0}$

c) d)

Section 2.2

2.2.1 : The function f' defined by the formula.

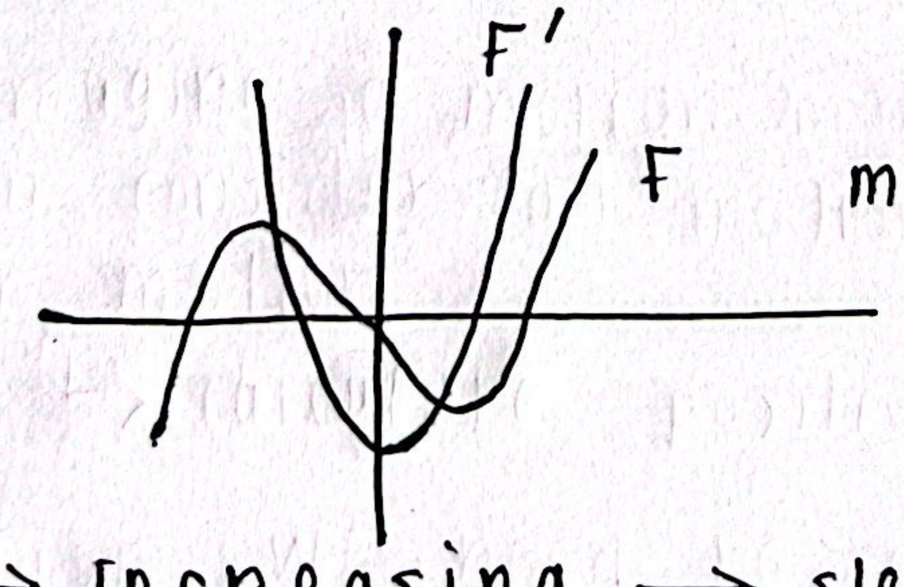
$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

$$= \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$$

* $F(x) = x^3 - x$

$F'(x) = 3x^2 - 1$



tan > 0 → Increasing → slope increase
→ Function increases

tan < 0 → Decreasing → slope decrease
→ Function decrease.

~~For F(x) The Funct~~

Home Works:

~~It~~ Do not need to solve exercise - 2.2.

Section 2.3 → need ~~to~~ study

Exercise set 2.3